

? If a cyclist is going around a curve at constant speed, is he accelerating? If so, what is the direction of his acceleration?

(i) No; (ii) yes, in the direction of his motion; (iii) yes, toward the inside of the curve; (iv) yes, toward the outside of the curve; (v) yes, but in some other direction.



3 Motion in Two or Three Dimensions

LEARNING OUTCOMES

In this chapter, you'll learn...

- 3.1 How to use vectors to represent the position and velocity of a particle in two or three dimensions.
- 3.2 How to find the vector acceleration of a particle, why a particle can have an acceleration even if its speed is constant, and how to interpret the components of acceleration parallel and perpendicular to a particle's path.
- 3.3 How to solve problems that involve the curved path followed by a projectile.
- 3.4 How to analyze motion in a circular path, with either constant speed or varying speed.
- 3.5 How to relate the velocities of a moving object as seen from two different frames of reference.

You'll need to review...

- 2.1 Average x-velocity.
- 2.2 Instantaneous x-velocity.
- 2.3 Average and instantaneous x-acceleration.
- 2.4 Straight-line motion with constant acceleration.
- 2.5 The motion of freely falling objects.

What determines where a batted baseball lands? How do you describe the motion of a roller coaster car along a curved track or the flight of a circling hawk? Which hits the ground first: a baseball that you simply drop or one that you throw horizontally?

We can't answer these kinds of questions by using the techniques of Chapter 2, in which particles moved only along a straight line. Instead, we need to extend our descriptions of motion to two- and three-dimensional situations. We'll still use the vector quantities displacement, velocity, and acceleration, but now these quantities will no longer lie along a single line. We'll find that several important kinds of motion take place in two dimensions only—that is, in a *plane*.

We also need to consider how the motion of a particle is described by different observers who are moving relative to each other. The concept of *relative velocity* will play an important role later in the book when we explore electromagnetic phenomena and when we introduce Einstein's special theory of relativity.

This chapter merges the vector mathematics of Chapter 1 with the kinematic language of Chapter 2. As before, we're concerned with describing motion, not with analyzing its causes. But the language you learn here will be an essential tool in later chapters when we study the relationship between force and motion.

3.1 POSITION AND VELOCITY VECTORS

Let's see how to describe a particle's motion in space. If the particle is at a point *P* at a certain instant, the **position vector** $\vec{r}$ of the particle at this instant is a vector that goes from the origin of the coordinate system to point *P* (Fig. 3.1). The Cartesian coordinates *x*, *y*, and *z* of point *P* are the *x*-, *y*-, and *z*-components of vector $\vec{r}$. Using the unit vectors we introduced in Section 1.9, we can write

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \quad (3.1)$$

Position vector of a particle at a given instant $\vec{r}$ Unit vectors in *x*-, *y*-, and *z*-directions $\hat{i}$, $\hat{j}$, $\hat{k}$ Coordinates of particle's position *x*, *y*, *z*

During a time interval Δt the particle moves from P_1 , where its position vector is $\vec{r}_1$, to P_2 , where its position vector is $\vec{r}_2$. The change in position (the displacement) during this interval is $\Delta\vec{r} = \vec{r}_2 - \vec{r}_1 = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$. We define the **average velocity** $\vec{v}_{av}$ during this interval in the same way we did in Chapter 2 for straight-line motion, as the displacement divided by the time interval (Fig. 3.2):

$$\vec{v}_{av} = \frac{\Delta\vec{r}}{\Delta t} = \frac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1} \quad (3.2)$$

Change in the particle's position vector
Final position minus initial position

Average velocity vector of a particle during time interval from t_1 to t_2
Time interval
Final time minus initial time

Dividing a vector by a scalar is a special case of *multiplying* a vector by a scalar, described in Section 1.7; the average velocity $\vec{v}_{av}$ is equal to the displacement vector $\Delta\vec{r}$ multiplied by $1/\Delta t$. Note that the x -component of Eq. (3.2) is $v_{av-x} = (x_2 - x_1)/(t_2 - t_1) = \Delta x/\Delta t$. This is just Eq. (2.2), the expression for average x -velocity that we found in Section 2.1 for one-dimensional motion.

We now define **instantaneous velocity** just as we did in Chapter 2: It equals the instantaneous rate of change of position with time. The key difference is that both position $\vec{r}$ and instantaneous velocity $\vec{v}$ are now vectors:

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{r}}{\Delta t} = \frac{d\vec{r}}{dt} \quad (3.3)$$

The instantaneous velocity vector of a particle ...
... equals the limit of its average velocity vector as the time interval approaches zero ...
... and equals the instantaneous rate of change of its position vector.

At any instant, the *magnitude* of $\vec{v}$ is the *speed* v of the particle at that instant, and the *direction* of $\vec{v}$ is the direction in which the particle is moving at that instant.

As $\Delta t \rightarrow 0$, points P_1 and P_2 in Fig. 3.2 move closer and closer together. In this limit, the vector $\Delta\vec{r}$ becomes tangent to the path. The direction of $\Delta\vec{r}$ in this limit is also the direction of $\vec{v}$. So *at every point along the path, the instantaneous velocity vector is tangent to the path at that point* (Fig. 3.3).

It's often easiest to calculate the instantaneous velocity vector by using components. During any displacement $\Delta\vec{r}$, the changes Δx , Δy , and Δz in the three coordinates of the particle are the *components* of $\Delta\vec{r}$. It follows that the components v_x , v_y , and v_z of the instantaneous velocity $\vec{v} = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$ are simply the time derivatives of the coordinates x , y , and z :

$$v_x = \frac{dx}{dt}, \quad v_y = \frac{dy}{dt}, \quad v_z = \frac{dz}{dt} \quad (3.4)$$

Each component of a particle's instantaneous velocity vector ...
... equals the instantaneous rate of change of its corresponding coordinate.

The x -component of $\vec{v}$ is $v_x = dx/dt$, which is the same as Eq. (2.3) for straight-line motion (see Section 2.2). Hence Eq. (3.4) is a direct extension of instantaneous velocity to motion in three dimensions.

We can also get Eq. (3.4) by taking the derivative of Eq. (3.1). The unit vectors $\hat{i}$, $\hat{j}$, and $\hat{k}$ don't depend on time, so their derivatives are zero and we find

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k} \quad (3.5)$$

This shows again that the components of $\vec{v}$ are dx/dt , dy/dt , and dz/dt .

The magnitude of the instantaneous velocity vector $\vec{v}$ —that is, the speed—is given in terms of the components v_x , v_y , and v_z by the Pythagorean relationship:

$$|\vec{v}| = v = \sqrt{v_x^2 + v_y^2 + v_z^2} \quad (3.6)$$

Figure 3.1 The position vector $\vec{r}$ from origin O to point P has components x , y , and z .

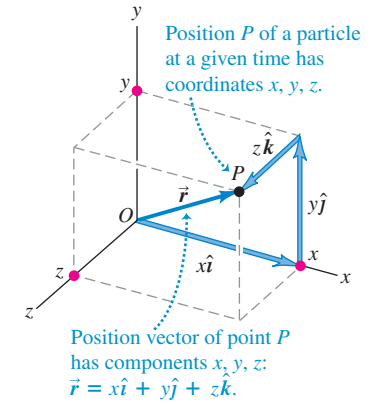


Figure 3.2 The average velocity $\vec{v}_{av}$ between points P_1 and P_2 has the same direction as the displacement $\Delta\vec{r}$.

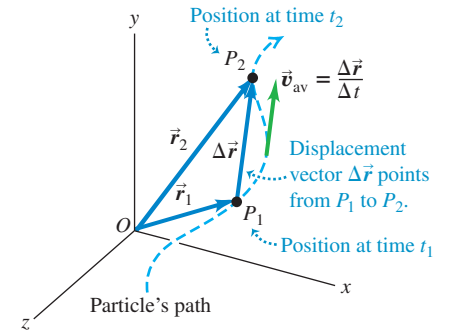


Figure 3.3 The vectors $\vec{v}_1$ and $\vec{v}_2$ are the instantaneous velocities at the points P_1 and P_2 shown in Fig. 3.2.

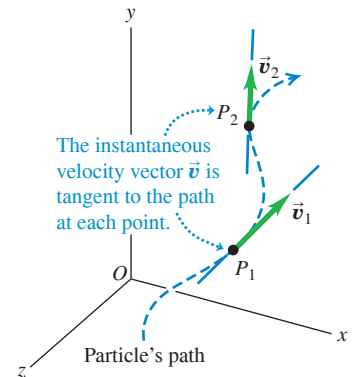


Figure 3.4 The two velocity components for motion in the xy -plane.

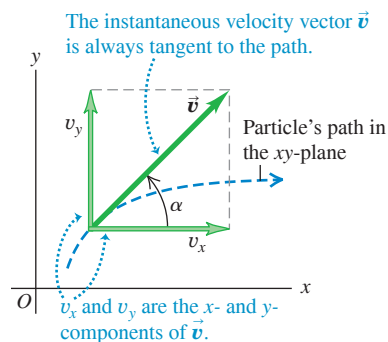


Figure 3.4 shows the situation when the particle moves in the xy -plane. In this case, z and v_z are zero. Then the speed (the magnitude of $\vec{v}$) is

$$v = \sqrt{v_x^2 + v_y^2}$$

and the direction of the instantaneous velocity $\vec{v}$ is given by angle α (the Greek letter alpha) in the figure. We see that

$$\tan \alpha = \frac{v_y}{v_x} \quad (3.7)$$

(We use α for the direction of the instantaneous velocity vector to avoid confusion with the direction θ of the *position* vector of the particle.)

From now on, when we use the word “velocity,” we’ll always mean the *instantaneous* velocity vector $\vec{v}$ (rather than the average velocity vector). Usually, we won’t even bother to call $\vec{v}$ a vector; it’s up to you to remember that velocity is a vector quantity with both magnitude and direction.

EXAMPLE 3.1 Calculating average and instantaneous velocity

A robotic vehicle, or rover, is exploring the surface of Mars. The stationary Mars lander is the origin of coordinates, and the surrounding Martian surface lies in the xy -plane. The rover, which we represent as a point, has x - and y -coordinates that vary with time:

$$\begin{aligned} x &= 2.0 \text{ m} - (0.25 \text{ m/s}^2)t^2 \\ y &= (1.0 \text{ m/s})t + (0.025 \text{ m/s}^3)t^3 \end{aligned}$$

(a) Find the rover’s coordinates and distance from the lander at $t = 2.0$ s. (b) Find the rover’s displacement and average velocity vectors for the interval $t = 0.0$ s to $t = 2.0$ s. (c) Find a general expression for the rover’s instantaneous velocity vector $\vec{v}$. Express $\vec{v}$ at $t = 2.0$ s in component form and in terms of magnitude and direction.

IDENTIFY and SET UP This problem involves motion in two dimensions, so we must use the vector equations obtained in this section. Figure 3.5 shows the rover’s path (dashed line). We’ll use Eq. (3.1) for position $\vec{r}$, the expression $\Delta \vec{r} = \vec{r}_2 - \vec{r}_1$ for displacement, Eq. (3.2) for average velocity, and Eqs. (3.5), (3.6), and (3.7) for instantaneous velocity and its magnitude and direction.

EXECUTE (a) At $t = 2.0$ s the rover’s coordinates are

$$\begin{aligned} x &= 2.0 \text{ m} - (0.25 \text{ m/s}^2)(2.0 \text{ s})^2 = 1.0 \text{ m} \\ y &= (1.0 \text{ m/s})(2.0 \text{ s}) + (0.025 \text{ m/s}^3)(2.0 \text{ s})^3 = 2.2 \text{ m} \end{aligned}$$

The rover’s distance from the origin at this time is

$$r = \sqrt{x^2 + y^2} = \sqrt{(1.0 \text{ m})^2 + (2.2 \text{ m})^2} = 2.4 \text{ m}$$

(b) To find the displacement and average velocity over the given time interval, we first express the position vector $\vec{r}$ as a function of time t . From Eq. (3.1) this is

$$\begin{aligned} \vec{r} &= x\hat{i} + y\hat{j} \\ &= [2.0 \text{ m} - (0.25 \text{ m/s}^2)t^2]\hat{i} + [(1.0 \text{ m/s})t + (0.025 \text{ m/s}^3)t^3]\hat{j} \end{aligned}$$

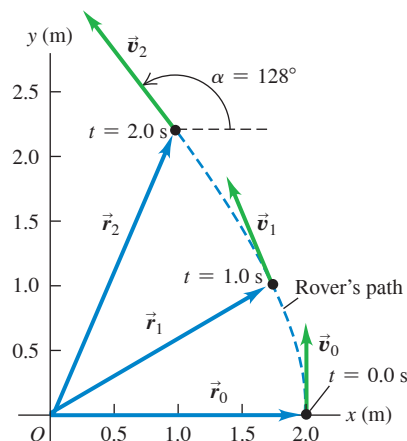
At $t = 0.0$ s the position vector $\vec{r}_0$ is

$$\vec{r}_0 = (2.0 \text{ m})\hat{i} + (0.0 \text{ m})\hat{j}$$

From part (a), the position vector $\vec{r}_2$ at $t = 2.0$ s is

$$\vec{r}_2 = (1.0 \text{ m})\hat{i} + (2.2 \text{ m})\hat{j}$$

Figure 3.5 At $t = 0.0$ s the rover has position vector $\vec{r}_0$ and instantaneous velocity vector $\vec{v}_0$. Likewise, $\vec{r}_1$ and $\vec{v}_1$ are the vectors at $t = 1.0$ s; $\vec{r}_2$ and $\vec{v}_2$ are the vectors at $t = 2.0$ s.



The displacement from $t = 0.0$ s to $t = 2.0$ s is therefore

$$\begin{aligned} \Delta \vec{r} &= \vec{r}_2 - \vec{r}_0 = (1.0 \text{ m})\hat{i} + (2.2 \text{ m})\hat{j} - (2.0 \text{ m})\hat{i} \\ &= (-1.0 \text{ m})\hat{i} + (2.2 \text{ m})\hat{j} \end{aligned}$$

During this interval the rover moves 1.0 m in the negative x -direction and 2.2 m in the positive y -direction. From Eq. (3.2), the average velocity over this interval is the displacement divided by the elapsed time:

$$\begin{aligned} \vec{v}_{\text{av}} &= \frac{\Delta \vec{r}}{\Delta t} = \frac{(-1.0 \text{ m})\hat{i} + (2.2 \text{ m})\hat{j}}{2.0 \text{ s} - 0.0 \text{ s}} \\ &= (-0.50 \text{ m/s})\hat{i} + (1.1 \text{ m/s})\hat{j} \end{aligned}$$

The components of this average velocity are $v_{\text{av-}x} = -0.50$ m/s and $v_{\text{av-}y} = 1.1$ m/s.

(c) From Eq. (3.4) the components of *instantaneous* velocity are the time derivatives of the coordinates:

$$\begin{aligned} v_x &= \frac{dx}{dt} = (-0.25 \text{ m/s}^2)(2t) \\ v_y &= \frac{dy}{dt} = 1.0 \text{ m/s} + (0.025 \text{ m/s}^3)(3t^2) \end{aligned}$$

Hence the instantaneous velocity vector is

$$\begin{aligned}\vec{v} &= v_x \hat{i} + v_y \hat{j} \\ &= (-0.50 \text{ m/s}^2)t \hat{i} + [1.0 \text{ m/s} + (0.075 \text{ m/s}^3)t^2] \hat{j}\end{aligned}$$

At $t = 2.0 \text{ s}$ the velocity vector $\vec{v}_2$ has components

$$\begin{aligned}v_{2x} &= (-0.50 \text{ m/s}^2)(2.0 \text{ s}) = -1.0 \text{ m/s} \\ v_{2y} &= 1.0 \text{ m/s} + (0.075 \text{ m/s}^3)(2.0 \text{ s})^2 = 1.3 \text{ m/s}\end{aligned}$$

The magnitude of the instantaneous velocity (that is, the speed) at $t = 2.0 \text{ s}$ is

$$v_2 = \sqrt{v_{2x}^2 + v_{2y}^2} = \sqrt{(-1.0 \text{ m/s})^2 + (1.3 \text{ m/s})^2} = 1.6 \text{ m/s}$$

Figure 3.5 shows the direction of velocity vector $\vec{v}_2$, which is at an angle α between 90° and 180° with respect to the positive x -axis. From Eq. (3.7) we have

$$\arctan \frac{v_y}{v_x} = \arctan \frac{1.3 \text{ m/s}}{-1.0 \text{ m/s}} = -52^\circ$$

This is off by 180° ; the correct value is $\alpha = 180^\circ - 52^\circ = 128^\circ$, or 38° west of north.

EVALUATE Compare the components of *average* velocity from part (b) for the interval from $t = 0.0 \text{ s}$ to $t = 2.0 \text{ s}$ ($v_{\text{av-}x} = -0.50 \text{ m/s}$, $v_{\text{av-}y} = 1.1 \text{ m/s}$) with the components of *instantaneous* velocity at $t = 2.0 \text{ s}$ from part (c) ($v_{2x} = -1.0 \text{ m/s}$, $v_{2y} = 1.3 \text{ m/s}$). Just as in one dimension, the average velocity vector $\vec{v}_{\text{av}}$ over an interval is in general *not* equal to the instantaneous velocity $\vec{v}$ at the end of the interval (see Example 2.1).

Figure 3.5 shows the position vectors $\vec{r}$ and instantaneous velocity vectors $\vec{v}$ at $t = 0.0 \text{ s}$, 1.0 s , and 2.0 s . (Calculate these quantities for $t = 0.0 \text{ s}$ and $t = 1.0 \text{ s}$.) Notice that $\vec{v}$ is tangent to the path at every point. The magnitude of $\vec{v}$ increases as the rover moves, which means that its speed is increasing.

KEYCONCEPT To calculate the average velocity vector of an object, first find its displacement vector during a time interval. Then divide by the time interval. To calculate the object's instantaneous velocity vector (its average velocity vector over an infinitesimally short time interval), take the derivative of its position vector with respect to time.

TEST YOUR UNDERSTANDING OF SECTION 3.1 In which of these situations *would* the average velocity vector $\vec{v}_{\text{av}}$ over an interval be equal to the instantaneous velocity $\vec{v}$ at the end of the interval? (i) An object moving along a curved path at constant speed; (ii) an object moving along a curved path and speeding up; (iii) an object moving along a straight line at constant speed; (iv) an object moving along a straight line and speeding up.

ANSWER

(iv) If the instantaneous velocity $\vec{v}$ is constant over an interval, its value at any point (including the end of the interval) is the same as the average velocity $\vec{v}_{\text{av}}$ over the interval. In (i) and (ii) the direction of $\vec{v}$ at the end of the interval is tangent to the path at that point, while the direction of $\vec{v}_{\text{av}}$ points from the beginning of the path to its end (in the direction of the net displacement). In (iii) both $\vec{v}$ and $\vec{v}_{\text{av}}$ are directed along the straight line, but $\vec{v}$ has a greater magnitude because speed has been increasing.

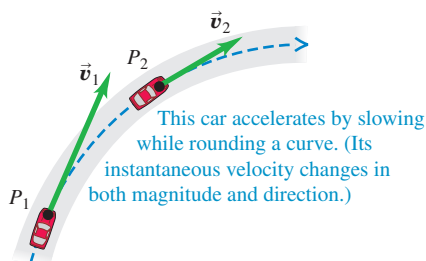
3.2 THE ACCELERATION VECTOR

Now let's consider the *acceleration* of a particle moving in space. Just as for motion in a straight line, acceleration describes how the velocity of the particle changes. But since we now treat velocity as a vector, acceleration will describe changes in the velocity magnitude (that is, the speed) *and* changes in the direction of velocity (that is, the direction in which the particle is moving).

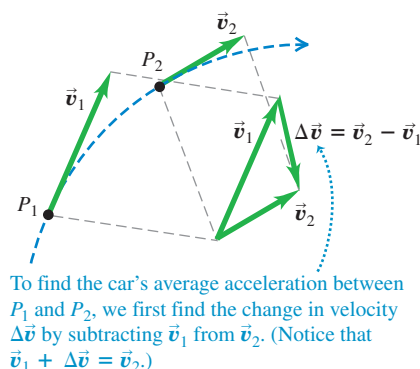
In **Fig. 3.6a**, a car (treated as a particle) is moving along a curved road. Vectors $\vec{v}_1$ and $\vec{v}_2$ represent the car's instantaneous velocities at time t_1 , when the car is at point

Figure 3.6 (a) A car moving along a curved road from P_1 to P_2 . (b) How to obtain the change in velocity $\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$ by vector subtraction. (c) The vector $\vec{a}_{\text{av}} = \Delta\vec{v}/\Delta t$ represents the average acceleration between P_1 and P_2 .

(a)



(b)



(c)

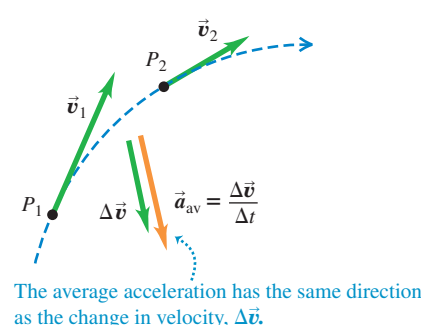
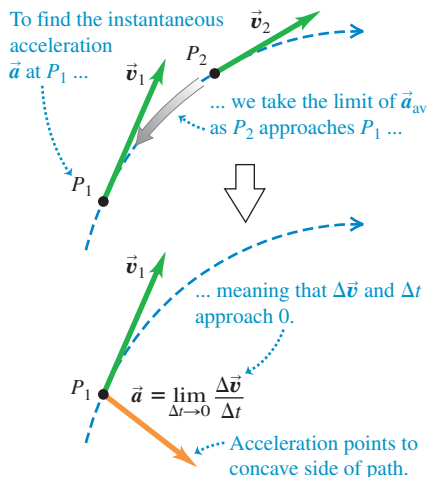
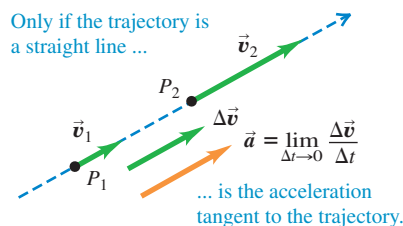


Figure 3.7 (a) Instantaneous acceleration $\vec{a}$ at point P_1 in Fig. 3.6. (b) Instantaneous acceleration for motion along a straight line.

(a) Acceleration: curved trajectory



(b) Acceleration: straight-line trajectory



BIO APPLICATION **Horses on a Curved Path** By leaning to the side and hitting the ground with their hooves at an angle, these horses give themselves the sideways acceleration necessary to make a sharp change in direction.



P_1 , and at time t_2 , when the car is at point P_2 . During the time interval from t_1 to t_2 , the **vector change in velocity** is $\vec{v}_2 - \vec{v}_1 = \Delta \vec{v}$, so $\vec{v}_2 = \vec{v}_1 + \Delta \vec{v}$ (Fig. 3.6b). The **average acceleration** $\vec{a}_{av}$ of the car during this time interval is the velocity change divided by the time interval $t_2 - t_1 = \Delta t$:

$$\vec{a}_{av} = \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} \quad (3.8)$$

Change in the particle's velocity

Average acceleration vector of a particle during time interval from t_1 to t_2

Time interval

Final velocity minus initial velocity

Final time minus initial time

Average acceleration is a **vector** quantity in the same direction as $\Delta \vec{v}$ (Fig. 3.6c). The x -component of Eq. (3.8) is $a_{av-x} = (v_{2x} - v_{1x})/(t_2 - t_1) = \Delta v_x/\Delta t$, which is just Eq. (2.4) for average acceleration in straight-line motion.

As in Chapter 2, we define the **instantaneous acceleration** $\vec{a}$ (a **vector** quantity) at point P_1 as the limit of the average acceleration vector when point P_2 approaches point P_1 , so both $\Delta \vec{v}$ and Δt approach zero (**Fig. 3.7**):

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} \quad (3.9)$$

The instantaneous acceleration vector of a particle ...

... equals the limit of its average acceleration vector as the time interval approaches zero ...

... and equals the instantaneous rate of change of its velocity vector.

The velocity vector $\vec{v}$ is always tangent to the particle's path, but the instantaneous acceleration vector $\vec{a}$ does *not* have to be tangent to the path. If the path is curved, $\vec{a}$ points toward the concave side of the path—that is, toward the inside of any turn that the particle is making (Fig. 3.7a). The acceleration is tangent to the path only if the particle moves in a straight line (Fig. 3.7b).

CAUTION Any particle following a curved path is accelerating When a particle is moving in a curved path, it always has nonzero acceleration, even when it moves with constant speed. This conclusion is contrary to the everyday use of the word “acceleration” to mean that speed is increasing. The more precise definition given in Eq. (3.9) shows that there is a nonzero acceleration whenever the velocity vector changes in *any* way, whether there is a change of speed, direction, or both. ■

To convince yourself that a particle is accelerating as it moves on a curved path with constant speed, think of your sensations when you ride in a car. When the car accelerates, you tend to move inside the car in a direction *opposite* to the car's acceleration. (In Chapter 4 we'll learn why this is so.) Thus you tend to slide toward the back of the car when it accelerates forward (speeds up) and toward the front of the car when it accelerates backward (slows down). If the car makes a turn on a level road, you tend to slide toward the outside of the turn; hence the car is accelerating toward the inside of the turn.

We'll usually be interested in instantaneous acceleration, not average acceleration. From now on, we'll use the term “acceleration” to mean the instantaneous acceleration vector $\vec{a}$.

Each component of the acceleration vector $\vec{a} = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$ is the derivative of the corresponding component of velocity:

$$a_x = \frac{dv_x}{dt} \quad a_y = \frac{dv_y}{dt} \quad a_z = \frac{dv_z}{dt} \quad (3.10)$$

Each component of a particle's instantaneous acceleration vector ...

... equals the instantaneous rate of change of its corresponding velocity component.

In terms of unit vectors,

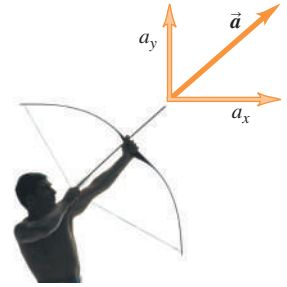
$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{dv_x}{dt}\hat{i} + \frac{dv_y}{dt}\hat{j} + \frac{dv_z}{dt}\hat{k} \quad (3.11)$$

The x -component of Eqs. (3.10) and (3.11), $a_x = dv_x/dt$, is just Eq. (2.5) for instantaneous acceleration in one dimension. **Figure 3.8** shows an example of an acceleration vector that has both x - and y -components.

Since each component of velocity is the derivative of the corresponding coordinate, we can express the components a_x , a_y , and a_z of the acceleration vector $\vec{a}$ as

$$a_x = \frac{d^2x}{dt^2} \quad a_y = \frac{d^2y}{dt^2} \quad a_z = \frac{d^2z}{dt^2} \quad (3.12)$$

Figure 3.8 When the fingers release the arrow, its acceleration vector has a horizontal component (a_x) and a vertical component (a_y).



EXAMPLE 3.2 Calculating average and instantaneous acceleration

Let's return to the motions of the Mars rover in Example 3.1. (a) Find the components of the average acceleration for the interval $t = 0.0$ s to $t = 2.0$ s. (b) Find the instantaneous acceleration at $t = 2.0$ s.

IDENTIFY and SET UP In Example 3.1 we found the components of the rover's instantaneous velocity at any time t :

$$v_x = \frac{dx}{dt} = (-0.25 \text{ m/s}^2)(2t) = (-0.50 \text{ m/s}^2)t$$

$$v_y = \frac{dy}{dt} = 1.0 \text{ m/s} + (0.025 \text{ m/s}^3)(3t^2) = 1.0 \text{ m/s} + (0.075 \text{ m/s}^3)t^2$$

We'll use the vector relationships among velocity, average acceleration, and instantaneous acceleration. In part (a) we determine the values of v_x and v_y at the beginning and end of the interval and then use Eq. (3.8) to calculate the components of the average acceleration. In part (b) we obtain expressions for the instantaneous acceleration components at any time t by taking the time derivatives of the velocity components as in Eqs. (3.10).

EXECUTE (a) In Example 3.1 we found that at $t = 0.0$ s the velocity components are

$$v_x = 0.0 \text{ m/s} \quad v_y = 1.0 \text{ m/s}$$

and that at $t = 2.0$ s the components are

$$v_x = -1.0 \text{ m/s} \quad v_y = 1.3 \text{ m/s}$$

Thus the components of average acceleration in the interval $t = 0.0$ s to $t = 2.0$ s are

$$a_{\text{av-}x} = \frac{\Delta v_x}{\Delta t} = \frac{-1.0 \text{ m/s} - 0.0 \text{ m/s}}{2.0 \text{ s} - 0.0 \text{ s}} = -0.50 \text{ m/s}^2$$

$$a_{\text{av-}y} = \frac{\Delta v_y}{\Delta t} = \frac{1.3 \text{ m/s} - 1.0 \text{ m/s}}{2.0 \text{ s} - 0.0 \text{ s}} = 0.15 \text{ m/s}^2$$

(b) Using Eqs. (3.10), we find

$$a_x = \frac{dv_x}{dt} = -0.50 \text{ m/s}^2 \quad a_y = \frac{dv_y}{dt} = (0.075 \text{ m/s}^3)(2t)$$

Hence the instantaneous acceleration vector $\vec{a}$ at time t is

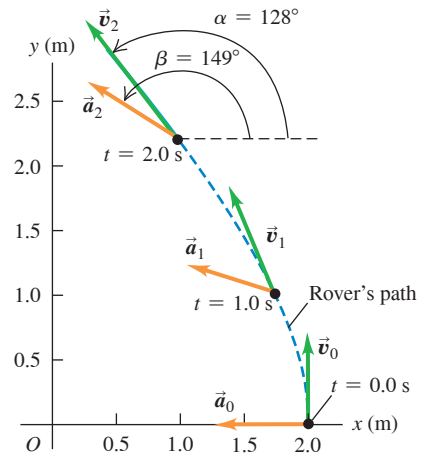
$$\vec{a} = a_x\hat{i} + a_y\hat{j} = (-0.50 \text{ m/s}^2)\hat{i} + (0.15 \text{ m/s}^3)t\hat{j}$$

At $t = 2.0$ s the components of acceleration and the acceleration vector are

$$a_x = -0.50 \text{ m/s}^2 \quad a_y = (0.15 \text{ m/s}^3)(2.0 \text{ s}) = 0.30 \text{ m/s}^2$$

$$\vec{a} = (-0.50 \text{ m/s}^2)\hat{i} + (0.30 \text{ m/s}^2)\hat{j}$$

Figure 3.9 The path of the robotic rover, showing the velocity and acceleration at $t = 0.0$ s ($\vec{v}_0$ and $\vec{a}_0$), $t = 1.0$ s ($\vec{v}_1$ and $\vec{a}_1$), and $t = 2.0$ s ($\vec{v}_2$ and $\vec{a}_2$).



The magnitude of acceleration at this time is

$$a = \sqrt{a_x^2 + a_y^2} = \sqrt{(-0.50 \text{ m/s}^2)^2 + (0.30 \text{ m/s}^2)^2} = 0.58 \text{ m/s}^2$$

A sketch of this vector (**Fig. 3.9**) shows that the direction angle β of $\vec{a}$ with respect to the positive x -axis is between 90° and 180° . From Eq. (3.7) we have

$$\arctan \frac{a_y}{a_x} = \arctan \frac{0.30 \text{ m/s}^2}{-0.50 \text{ m/s}^2} = -31^\circ$$

Hence $\beta = 180^\circ + (-31^\circ) = 149^\circ$.

EVALUATE Figure 3.9 shows the rover's path and the velocity and acceleration vectors at $t = 0.0$ s, 1.0 s, and 2.0 s. (Use the results of part (b) to calculate the instantaneous acceleration at $t = 0.0$ s and $t = 1.0$ s for yourself.) Note that $\vec{v}$ and $\vec{a}$ are *not* in the same direction at any of these times. The velocity vector $\vec{v}$ is tangent to the path at each point (as is always the case), and the acceleration vector $\vec{a}$ points toward the concave side of the path.

KEYCONCEPT To calculate the average acceleration vector of an object, first find the change in its velocity vector (final velocity minus initial velocity) during a time interval. Then divide by the time interval. To calculate the object's instantaneous acceleration vector (its average velocity vector over an infinitesimally short time interval), take the derivative of its velocity vector with respect to time.

Figure 3.10 The acceleration can be resolved into a component $a_{\parallel}$ parallel to the path (that is, along the tangent to the path) and a component $a_{\perp}$ perpendicular to the path (that is, along the normal to the path).

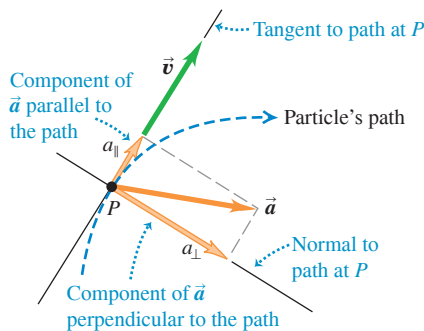
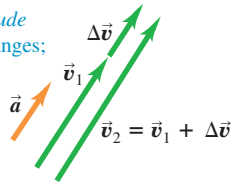


Figure 3.11 The effect of acceleration directed (a) parallel to and (b) perpendicular to a particle's velocity.

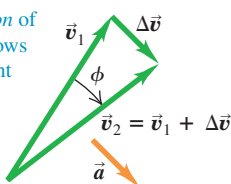
(a) Acceleration parallel to velocity

Changes only *magnitude* of velocity: speed changes; direction doesn't.



(b) Acceleration perpendicular to velocity

Changes only *direction* of velocity: particle follows curved path at constant speed.



Parallel and Perpendicular Components of Acceleration

Equations (3.10) tell us about the components of a particle's instantaneous acceleration vector $\vec{a}$ along the x -, y -, and z -axes. Another useful way to think about $\vec{a}$ is in terms of one component *parallel* to the particle's path and to its velocity $\vec{v}$, and one component *perpendicular* to the path and to $\vec{v}$ (Fig. 3.10). That's because the parallel component $a_{\parallel}$ tells us about changes in the particle's *speed*, while the perpendicular component $a_{\perp}$ tells us about changes in the particle's *direction of motion*. To see why the parallel and perpendicular components of $\vec{a}$ have these properties, let's consider two special cases.

In Fig. 3.11a the acceleration vector is in the same direction as the velocity $\vec{v}_1$, so $\vec{a}$ has only a parallel component $a_{\parallel}$ (that is, $a_{\perp} = 0$). The velocity change $\Delta\vec{v}$ during a small time interval Δt is in the same direction as $\vec{a}$ and hence in the same direction as $\vec{v}_1$. The velocity $\vec{v}_2$ at the end of Δt is in the same direction as $\vec{v}_1$ but has greater magnitude. Hence during the time interval Δt the particle in Fig. 3.11a moved in a straight line with increasing speed (compare Fig. 3.7b).

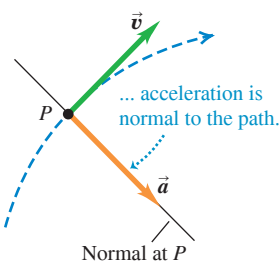
In Fig. 3.11b the acceleration is *perpendicular* to the velocity, so $\vec{a}$ has only a perpendicular component $a_{\perp}$ (that is, $a_{\parallel} = 0$). In a small time interval Δt , the velocity change $\Delta\vec{v}$ is very nearly perpendicular to $\vec{v}_1$, and so $\vec{v}_1$ and $\vec{v}_2$ have different directions. As the time interval Δt approaches zero, the angle ϕ in the figure also approaches zero, $\Delta\vec{v}$ becomes perpendicular to *both* $\vec{v}_1$ and $\vec{v}_2$, and $\vec{v}_1$ and $\vec{v}_2$ have the same magnitude. In other words, the speed of the particle stays the same, but the direction of motion changes and the path of the particle curves.

In the most general case, the acceleration $\vec{a}$ has *both* components parallel and perpendicular to the velocity $\vec{v}$, as in Fig. 3.10. Then the particle's speed will change (described by the parallel component $a_{\parallel}$) and its direction of motion will change (described by the perpendicular component $a_{\perp}$).

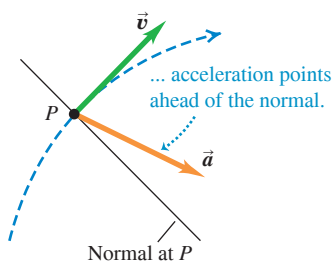
Figure 3.12 shows a particle moving along a curved path for three situations: constant speed, increasing speed, and decreasing speed. If the speed is constant, $\vec{a}$ is perpendicular, or *normal*, to the path and to $\vec{v}$ and points toward the concave side of the path (Fig. 3.12a). If the speed is increasing, there is still a perpendicular component of $\vec{a}$, but there is also a parallel component with the same direction as $\vec{v}$ (Fig. 3.12b). Then $\vec{a}$ points ahead of the normal to the path. (This was the case in Example 3.2.) If the speed is decreasing, the parallel component has the direction opposite to $\vec{v}$, and $\vec{a}$ points behind the normal to the path (Fig. 3.12c; compare Fig. 3.7a). We'll use these ideas again in Section 3.4 when we study the special case of motion in a circle.

Figure 3.12 Velocity and acceleration vectors for a particle moving through a point P on a curved path with (a) constant speed, (b) increasing speed, and (c) decreasing speed.

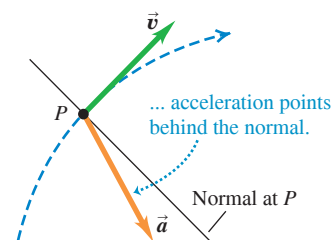
(a) When speed is constant along a curved path ...



(b) When speed is increasing along a curved path ...



(c) When speed is decreasing along a curved path ...



EXAMPLE 3.3 Calculating parallel and perpendicular components of acceleration

For the rover of Examples 3.1 and 3.2, find the parallel and perpendicular components of the acceleration at $t = 2.0$ s.

IDENTIFY and SET UP We want to find the components of the acceleration vector $\vec{a}$ that are parallel and perpendicular to velocity vector $\vec{v}$. We found the directions of $\vec{v}$ and $\vec{a}$ in Examples 3.1 and 3.2, respectively; Fig. 3.9 shows the results. From these directions we can find the angle between the two vectors and the components of $\vec{a}$ with respect to the direction of $\vec{v}$.

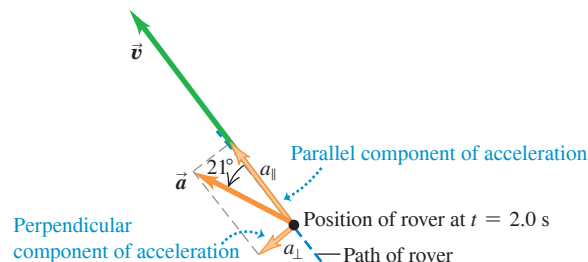
EXECUTE From Example 3.2, at $t = 2.0$ s the particle has an acceleration of magnitude 0.58 m/s^2 at an angle of 149° with respect to the positive x -axis. In Example 3.1 we found that at this time the velocity vector is at an angle of 128° with respect to the positive x -axis. The angle between $\vec{a}$ and $\vec{v}$ is therefore $149^\circ - 128^\circ = 21^\circ$ (Fig. 3.13). Hence the components of acceleration parallel and perpendicular to $\vec{v}$ are

$$a_{\parallel} = a \cos 21^\circ = (0.58 \text{ m/s}^2) \cos 21^\circ = 0.54 \text{ m/s}^2$$

$$a_{\perp} = a \sin 21^\circ = (0.58 \text{ m/s}^2) \sin 21^\circ = 0.21 \text{ m/s}^2$$

EVALUATE The parallel component $a_{\parallel}$ is positive (in the same direction as $\vec{v}$), which means that the speed is increasing at this instant. The value $a_{\parallel} = +0.54 \text{ m/s}^2$ tells us that the speed is increasing at this instant at a rate of 0.54 m/s per second. The perpendicular component $a_{\perp}$ is not

Figure 3.13 The parallel and perpendicular components of the acceleration of the rover at $t = 2.0$ s.



zero, which means that at this instant the rover is turning—that is, it is changing direction and following a curved path.

KEYCONCEPT If an object's *speed* is changing, there is a component of its acceleration vector *parallel* to its velocity vector. If an object's *direction of motion* is changing—that is, it is turning—there is a component of its acceleration vector *perpendicular* to its velocity vector and toward the inside of the turn.

CONCEPTUAL EXAMPLE 3.4 Acceleration of a skier

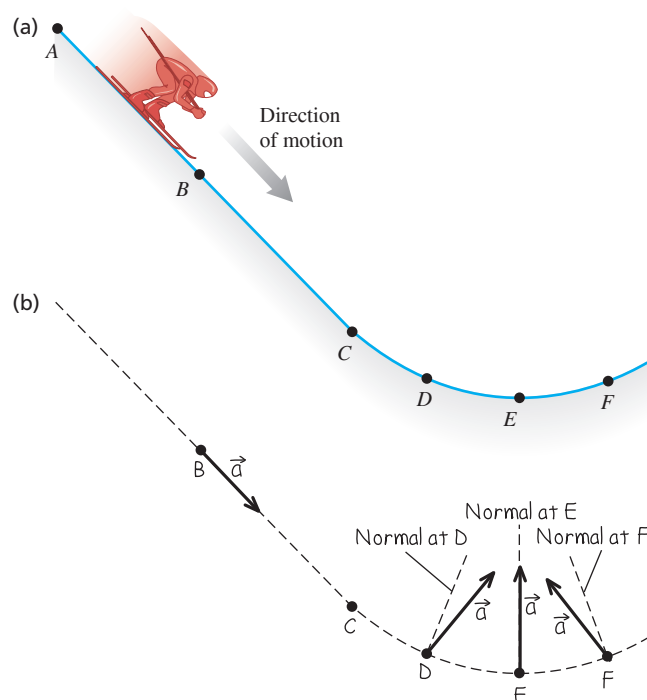
A skier moves along a ski-jump ramp (Fig. 3.14a). The ramp is straight from point A to point C and curved from point C onward. The skier speeds up as she moves downhill from point A to point E, where her speed is maximum. She slows down after passing point E. Draw the direction of the acceleration vector at each of the points B, D, E, and F.

SOLUTION Figure 3.14b shows our solution. At point B the skier is moving in a straight line with increasing speed, so her acceleration points downhill, in the same direction as her velocity. At points D, E, and F the skier is moving along a curved path, so her acceleration has a component perpendicular to the path (toward the concave side of the path) at each of these points. At point D there is also an acceleration component in the direction of her motion because she is speeding up. So the acceleration vector points *ahead* of the normal to her path at point D. At point E, the skier's speed is instantaneously not changing; her speed is maximum at this point, so its derivative is zero. There is therefore no parallel component of $\vec{a}$, and the acceleration is perpendicular to her motion. At point F there is an acceleration component *opposite* to the direction of her motion because she's slowing down. The acceleration vector therefore points *behind* the normal to her path.

In the next section we'll consider the skier's acceleration after she flies off the ramp.

KEYCONCEPT If a moving object is turning (changing direction), its acceleration vector points ahead of the normal to its path if it is speeding up, behind the normal if it is slowing down, and along the normal if its speed is instantaneously not changing.

Figure 3.14 (a) The skier's path. (b) Our solution.



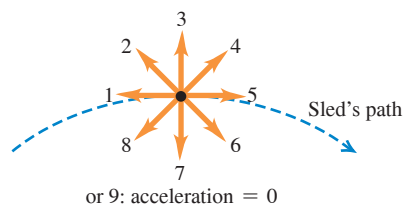


Figure 3.15 The trajectory of an idealized projectile.

- A projectile moves in a vertical plane that contains the initial velocity vector $\vec{v}_0$.
- Its trajectory depends only on $\vec{v}_0$ and on the downward acceleration due to gravity.

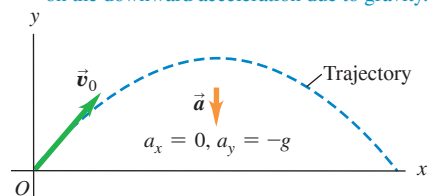
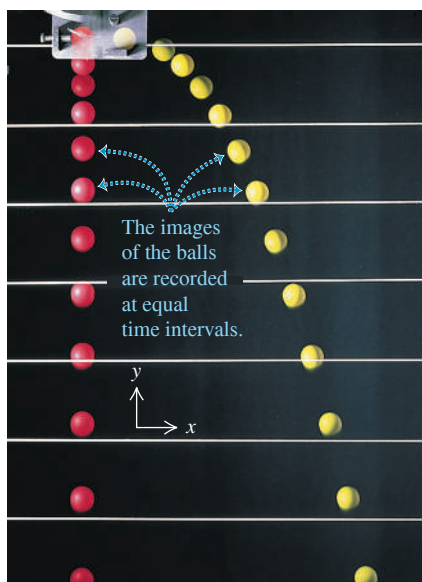


Figure 3.16 The red ball is dropped from rest, and the yellow ball is simultaneously projected horizontally.



- At any time the two balls have different x -coordinates and x -velocities but the same y -coordinate, y -velocity, and y -acceleration.
- The horizontal motion of the yellow ball has no effect on its vertical motion.

TEST YOUR UNDERSTANDING OF SECTION 3.2 A sled travels over the crest of a snow-covered hill. The sled slows down as it climbs up one side of the hill and gains speed as it descends on the other side. Which of the vectors (1 through 9) in the figure correctly shows the direction of the sled's acceleration at the crest? (Choice 9 is that the acceleration is zero.)

ANSWER

Vector 7 At the high point of the sled's path, the speed is minimum. At that point the speed is neither increasing nor decreasing, and the parallel component of the acceleration (that is, the horizontal component) is zero. The acceleration has only a perpendicular component toward the inside of the sled's curved path. In other words, the acceleration is downward.

3.3 PROJECTILE MOTION

A **projectile** is any object that is given an initial velocity and then follows a path determined entirely by the effects of gravitational acceleration and air resistance. A batted baseball, a thrown football, and a bullet shot from a rifle are all projectiles. The path followed by a projectile is called its **trajectory**.

To analyze the motion of a projectile, we'll use an idealized model. We'll represent the projectile as a particle with an acceleration (due to gravity) that is constant in both magnitude and direction. We'll ignore the effects of air resistance and the curvature and rotation of the earth. This model has limitations, however: We have to consider the earth's curvature when we study the flight of long-range missiles, and air resistance is of crucial importance to a sky diver. Nevertheless, we can learn a lot from analysis of this simple model. For the remainder of this chapter the phrase "projectile motion" will imply that we're ignoring air resistance. In Chapter 5 we'll see what happens when air resistance cannot be ignored.

Projectile motion is always confined to a vertical plane determined by the direction of the initial velocity (**Fig. 3.15**). This is because the acceleration due to gravity is purely vertical; gravity can't accelerate the projectile sideways. Thus projectile motion is *two-dimensional*. We'll call the plane of motion the xy -coordinate plane, with the x -axis horizontal and the y -axis vertically upward.

The key to analyzing projectile motion is that we can treat the x - and y -coordinates separately. **Figure 3.16** illustrates this for two projectiles: a red ball dropped from rest and a yellow ball projected horizontally from the same height. The figure shows that the horizontal motion of the yellow projectile has *no* effect on its vertical motion. For both projectiles, the x -component of acceleration is zero and the y -component is constant and equal to $-g$. (By definition, g is always positive; with our choice of coordinate directions, a_y is negative.) So we can analyze projectile motion as a combination of horizontal motion with constant velocity and vertical motion with constant acceleration.

We can then express all the vector relationships for the projectile's position, velocity, and acceleration by separate equations for the horizontal and vertical components. The components of $\vec{a}$ are

$$a_x = 0 \quad a_y = -g \quad (\text{projectile motion, no air resistance}) \quad (3.13)$$

Since both the x -acceleration and y -acceleration are constant, we can use Eqs. (2.8), (2.12), (2.13), and (2.14) directly. Suppose that at time $t = 0$ our particle is at the point (x_0, y_0) and its initial velocity at this time has components v_{0x} and v_{0y} . The components of acceleration are $a_x = 0$, $a_y = -g$. Considering the x -motion first, we substitute 0 for a_x in Eqs. (2.8) and (2.12). We find

$$v_x = v_{0x} \quad (3.14)$$

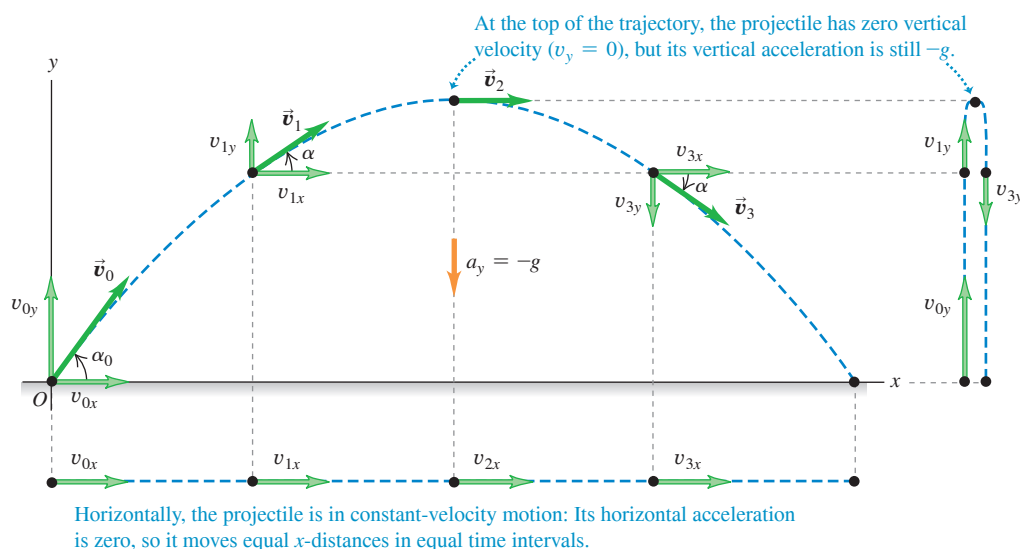
$$x = x_0 + v_{0x}t \quad (3.15)$$

For the y -motion we substitute y for x , v_y for v_x , v_{0y} for v_{0x} , and $a_y = -g$ for a_x :

$$v_y = v_{0y} - gt \quad (3.16)$$

$$y = y_0 + v_{0y}t - \frac{1}{2}gt^2 \quad (3.17)$$

Figure 3.17 If air resistance is negligible, the trajectory of a projectile is a combination of horizontal motion with constant velocity and vertical motion with constant acceleration.



It's usually simplest to take the initial position (at $t = 0$) as the origin; then $x_0 = y_0 = 0$. This might be the position of a ball at the instant it leaves the hand of the person who throws it or the position of a bullet at the instant it leaves the gun barrel.

Figure 3.17 shows the trajectory of a projectile that starts at (or passes through) the origin at time $t = 0$, along with its position, velocity, and velocity components at equal time intervals. The x -velocity v_x is constant; the y -velocity v_y changes by equal amounts in equal times, just as if the projectile were launched vertically with the same initial y -velocity.

We can also represent the initial velocity $\vec{v}_0$ by its magnitude v_0 (the initial speed) and its angle α_0 with the positive x -axis (**Fig. 3.18**). In terms of these quantities, the components v_{0x} and v_{0y} of the initial velocity are

$$v_{0x} = v_0 \cos \alpha_0 \quad v_{0y} = v_0 \sin \alpha_0 \quad (3.18)$$

If we substitute Eqs. (3.18) into Eqs. (3.14) through (3.17) and set $x_0 = y_0 = 0$, we get the following equations. They describe the position and velocity of the projectile in **Fig. 3.17** at any time t :

$$x = (v_0 \cos \alpha_0)t \quad (3.19)$$

$$y = (v_0 \sin \alpha_0)t - \frac{1}{2}gt^2 \quad (3.20)$$

$$v_x = v_0 \cos \alpha_0 \quad (3.21)$$

$$v_y = v_0 \sin \alpha_0 - gt \quad (3.22)$$

Figure 3.18 The initial velocity components v_{0x} and v_{0y} of a projectile (such as a kicked soccer ball) are related to the initial speed v_0 and initial angle α_0 .

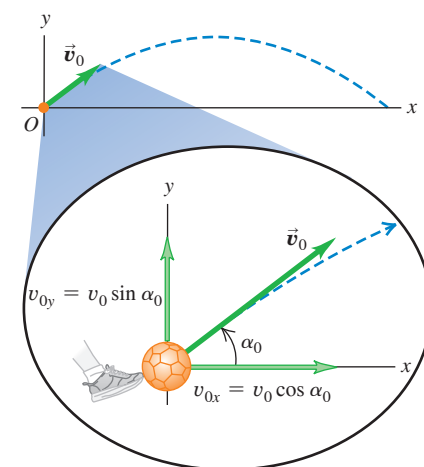


Figure 3.19 The nearly parabolic trajectories of a bouncing ball.

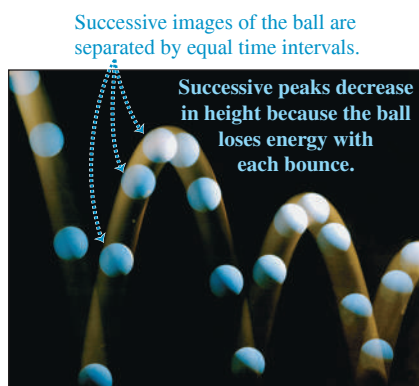
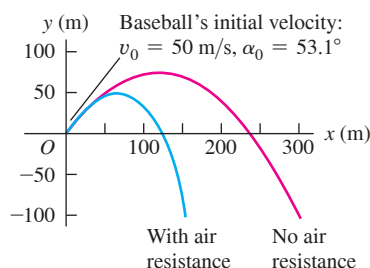


Figure 3.20 Air resistance has a large cumulative effect on the motion of a baseball. In this simulation we allow the baseball to fall below the height from which it was thrown (for example, the baseball could have been thrown from a cliff).



We can get a lot of information from Eqs. (3.19) through (3.22). For example, the distance r from the origin to the projectile at any time t is

$$r = \sqrt{x^2 + y^2} \quad (3.23)$$

The projectile's speed (the magnitude of its velocity) at any time is

$$v = \sqrt{v_x^2 + v_y^2} \quad (3.24)$$

The *direction* of the velocity, in terms of the angle α it makes with the positive x -direction (see Fig. 3.17), is

$$\tan \alpha = \frac{v_y}{v_x} \quad (3.25)$$

The velocity vector $\vec{v}$ is tangent to the trajectory at each point.

We can derive an equation for the trajectory's shape in terms of x and y by eliminating t . From Eqs. (3.19) and (3.20), we find $t = x/(v_0 \cos \alpha_0)$ and

$$y = (\tan \alpha_0)x - \frac{g}{2v_0^2 \cos^2 \alpha_0} x^2 \quad (3.26)$$

Don't worry about the details of this equation; the important point is its general form. Since v_0 , $\tan \alpha_0$, $\cos \alpha_0$, and g are constants, Eq. (3.26) has the form

$$y = bx - cx^2$$

where b and c are constants. This is the equation of a *parabola*. In our simple model of projectile motion, the trajectory is always a parabola (Fig. 3.19).

When air resistance *isn't* negligible and has to be included, calculating the trajectory becomes a lot more complicated; the effects of air resistance depend on velocity, so the acceleration is no longer constant. **Figure 3.20** shows a computer simulation of the trajectory of a baseball both without air resistance and with air resistance proportional to the square of the baseball's speed. We see that air resistance has a very large effect; the projectile does not travel as far or as high, and the trajectory is no longer a parabola.

CONCEPTUAL EXAMPLE 3.5 Acceleration of a skier, continued

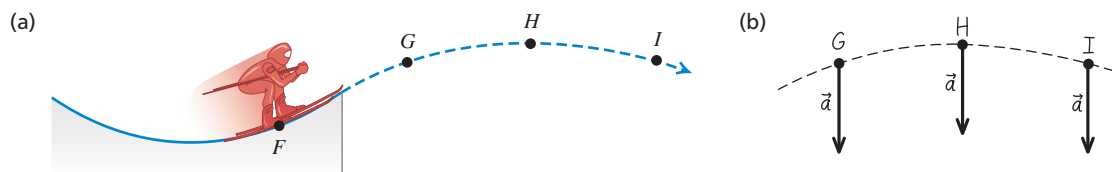
Let's consider again the skier in Conceptual Example 3.4. What is her acceleration at each of the points G , H , and I in **Fig. 3.21a** after she flies off the ramp? Neglect air resistance.

SOLUTION Figure 3.21b shows our answer. The skier's acceleration changed from point to point while she was on the ramp. But as soon as she leaves the ramp, she becomes a projectile. So at points G , H , and I ,

and indeed at *all* points after she leaves the ramp, the skier's acceleration points vertically downward and has magnitude g .

KEYCONCEPT No matter how complicated the acceleration of a particle before it becomes a projectile, its acceleration as a projectile is given by $a_x = 0$, $a_y = -g$.

Figure 3.21 (a) The skier's path during the jump. (b) Our solution.



PROBLEM-SOLVING STRATEGY 3.1 Projectile Motion

NOTE: The strategies we used in Sections 2.4 and 2.5 for straight-line, constant-acceleration problems are also useful here.

IDENTIFY *the relevant concepts:* The key concept is that throughout projectile motion, the acceleration is downward and has a constant magnitude g . Projectile-motion equations don't apply to *throwing* a ball, because during the throw the ball is acted on by both the thrower's hand and gravity. These equations apply only *after* the ball leaves the thrower's hand.

SET UP *the problem* using the following steps:

1. Define your coordinate system and make a sketch showing your axes. It's almost always best to make the x -axis horizontal and the y -axis vertical, and to choose the origin to be where the object first becomes a projectile (for example, where a ball leaves the thrower's hand). Then the components of acceleration are $a_x = 0$ and $a_y = -g$, as in Eq. (3.13); the initial position is $x_0 = y_0 = 0$; and you can use Eqs. (3.19) through (3.22). (If you choose a different origin or axes, you'll have to modify these equations.)
2. List the unknown and known quantities, and decide which unknowns are your target variables. For example, you might be given the initial velocity (either the components or the magnitude and direction) and asked to find the coordinates and velocity components

at some later time. Make sure that you have as many equations as there are target variables to be found. In addition to Eqs. (3.19) through (3.22), Eqs. (3.23) through (3.26) may be useful.

3. State the problem in words and then translate those words into symbols. For example, *when* does the particle arrive at a certain point? (That is, at what value of t ?) *Where* is the particle when its velocity has a certain value? (That is, what are the values of x and y when v_x or v_y has the specified value?) Since $v_y = 0$ at the highest point in a trajectory, the question "When does the projectile reach its highest point?" translates into "What is the value of t when $v_y = 0$?" Similarly, "When does the projectile return to its initial elevation?" translates into "What is the value of t when $y = y_0$?"

EXECUTE *the solution:* Find the target variables using the equations you chose. Resist the temptation to break the trajectory into segments and analyze each segment separately. You don't have to start all over when the projectile reaches its highest point! It's almost always easier to use the same axes and time scale throughout the problem. If you need numerical values, use $g = 9.80 \text{ m/s}^2$. Remember that g is positive!

EVALUATE *your answer:* Do your results make sense? Do the numerical values seem reasonable?

EXAMPLE 3.6 An object projected horizontally

WITH  **ARIATION PROBLEMS**

A motorcycle stunt rider rides off the edge of a cliff. Just at the edge his velocity is horizontal, with magnitude 9.0 m/s . Find the motorcycle's position, distance from the edge of the cliff, and velocity 0.50 s after it leaves the edge of the cliff. Ignore air resistance.

IDENTIFY and SET UP Figure 3.22 shows our sketch of the trajectory of motorcycle and rider. He is in projectile motion as soon as he leaves the edge of the cliff, which we take to be the origin (so $x_0 = y_0 = 0$). His initial velocity $\vec{v}_0$ at the edge of the cliff is horizontal (that is, $\alpha_0 = 0$), so its components are $v_{0x} = v_0 \cos \alpha_0 = 9.0 \text{ m/s}$ and $v_{0y} = v_0 \sin \alpha_0 = 0$. To find the motorcycle's position at $t = 0.50 \text{ s}$, we use Eqs. (3.19) and (3.20); we then find the distance from the origin using Eq. (3.23). Finally, we use Eqs. (3.21) and (3.22) to find the velocity components at $t = 0.50 \text{ s}$.

EXECUTE From Eqs. (3.19) and (3.20), the motorcycle's x - and y -coordinates at $t = 0.50 \text{ s}$ are

$$x = v_{0x}t = (9.0 \text{ m/s})(0.50 \text{ s}) = 4.5 \text{ m}$$

$$y = -\frac{1}{2}gt^2 = -\frac{1}{2}(9.80 \text{ m/s}^2)(0.50 \text{ s})^2 = -1.2 \text{ m}$$

The negative value of y shows that the motorcycle is below its starting point.

From Eq. (3.23), the motorcycle's distance from the origin at $t = 0.50 \text{ s}$ is

$$r = \sqrt{x^2 + y^2} = \sqrt{(4.5 \text{ m})^2 + (-1.2 \text{ m})^2} = 4.7 \text{ m}$$

From Eqs. (3.21) and (3.22), the velocity components at $t = 0.50 \text{ s}$ are

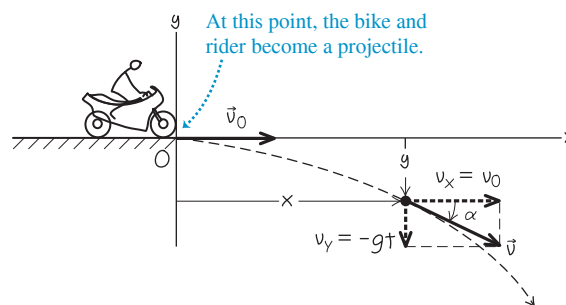
$$v_x = v_{0x} = 9.0 \text{ m/s}$$

$$v_y = -gt = (-9.80 \text{ m/s}^2)(0.50 \text{ s}) = -4.9 \text{ m/s}$$

The motorcycle has the same horizontal velocity v_x as when it left the cliff at $t = 0$, but in addition there is a downward (negative) vertical velocity v_y . The velocity vector at $t = 0.50 \text{ s}$ is

$$\vec{v} = v_x\hat{i} + v_y\hat{j} = (9.0 \text{ m/s})\hat{i} + (-4.9 \text{ m/s})\hat{j}$$

Figure 3.22 Our sketch for this problem.



From Eqs. (3.24) and (3.25), at $t = 0.50 \text{ s}$ the velocity has magnitude v and angle α given by

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{(9.0 \text{ m/s})^2 + (-4.9 \text{ m/s})^2} = 10.2 \text{ m/s}$$

$$\alpha = \arctan \frac{v_y}{v_x} = \arctan \left(\frac{-4.9 \text{ m/s}}{9.0 \text{ m/s}} \right) = -29^\circ$$

The motorcycle is moving at 10.2 m/s in a direction 29° below the horizontal.

EVALUATE Just as in Fig. 3.17, the motorcycle's horizontal motion is unchanged by gravity; the motorcycle continues to move horizontally at 9.0 m/s , covering 4.5 m in 0.50 s . The motorcycle initially has zero vertical velocity, so it falls vertically just like an object released from rest and descends a distance $\frac{1}{2}gt^2 = 1.2 \text{ m}$ in 0.50 s .

KEYCONCEPT The motion of a projectile is a combination of motion with constant velocity in the horizontal x -direction and motion with constant downward acceleration in the vertical y -direction.

EXAMPLE 3.7 Height and range of a projectile I: A batted baseball**WITH VARIATION PROBLEMS**

A batter hits a baseball so that it leaves the bat at speed $v_0 = 37.0$ m/s at an angle $\alpha_0 = 53.1^\circ$. (a) Find the position of the ball and its velocity (magnitude and direction) at $t = 2.00$ s. (b) Find the time when the ball reaches the highest point of its flight, and its height h at this time. (c) Find the *horizontal range* R —that is, the horizontal distance from the starting point to where the ball hits the ground—and the ball's velocity just before it hits.

IDENTIFY and SET UP As Fig. 3.20 shows, air resistance strongly affects the motion of a baseball. For simplicity, however, we'll ignore air resistance here and use the projectile-motion equations to describe the motion. The ball leaves the bat at $t = 0$ a meter or so above ground level, but we'll ignore this distance and assume that it starts at ground level ($y_0 = 0$). **Figure 3.23** shows our sketch of the ball's trajectory. We'll use the same coordinate system as in Figs. 3.17 and 3.18, so we can use Eqs. (3.19) through (3.22). Our target variables are (a) the position and velocity of the ball 2.00 s after it leaves the bat, (b) the time t when the ball is at its maximum height (that is, when $v_y = 0$) and the y -coordinate at this time, and (c) the x -coordinate when the ball returns to ground level ($y = 0$) and the ball's vertical component of velocity then.

EXECUTE (a) We want to find x , y , v_x , and v_y at $t = 2.00$ s. The initial velocity of the ball has components

$$\begin{aligned}v_{0x} &= v_0 \cos \alpha_0 = (37.0 \text{ m/s}) \cos 53.1^\circ = 22.2 \text{ m/s} \\v_{0y} &= v_0 \sin \alpha_0 = (37.0 \text{ m/s}) \sin 53.1^\circ = 29.6 \text{ m/s}\end{aligned}$$

From Eqs. (3.19) through (3.22),

$$\begin{aligned}x &= v_{0x}t = (22.2 \text{ m/s})(2.00 \text{ s}) = 44.4 \text{ m} \\y &= v_{0y}t - \frac{1}{2}gt^2 \\&= (29.6 \text{ m/s})(2.00 \text{ s}) - \frac{1}{2}(9.80 \text{ m/s}^2)(2.00 \text{ s})^2 = 39.6 \text{ m} \\v_x &= v_{0x} = 22.2 \text{ m/s} \\v_y &= v_{0y} - gt = 29.6 \text{ m/s} - (9.80 \text{ m/s}^2)(2.00 \text{ s}) = 10.0 \text{ m/s}\end{aligned}$$

The y -component of velocity is positive at $t = 2.00$ s, so the ball is still moving upward (Fig. 3.23). From Eqs. (3.24) and (3.25), the magnitude and direction of the velocity are

$$\begin{aligned}v &= \sqrt{v_x^2 + v_y^2} = \sqrt{(22.2 \text{ m/s})^2 + (10.0 \text{ m/s})^2} = 24.4 \text{ m/s} \\ \alpha &= \arctan\left(\frac{10.0 \text{ m/s}}{22.2 \text{ m/s}}\right) = \arctan 0.450 = 24.2^\circ\end{aligned}$$

The ball is moving at 24.4 m/s in a direction 24.2° above the horizontal.

(b) At the highest point, the vertical velocity v_y is zero. Call the time when this happens t_1 ; then

$$\begin{aligned}v_y &= v_{0y} - gt_1 = 0 \\t_1 &= \frac{v_{0y}}{g} = \frac{29.6 \text{ m/s}}{9.80 \text{ m/s}^2} = 3.02 \text{ s}\end{aligned}$$

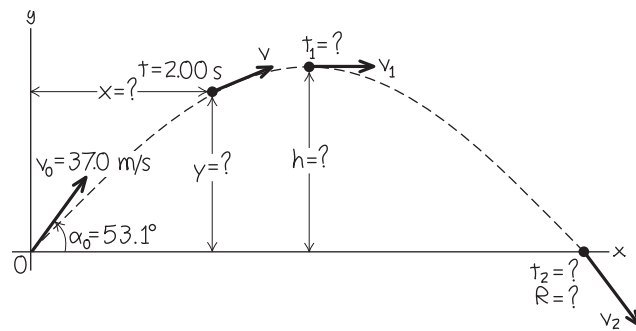
The height h at the highest point is the value of y at time t_1 :

$$\begin{aligned}h &= v_{0y}t_1 - \frac{1}{2}gt_1^2 \\&= (29.6 \text{ m/s})(3.02 \text{ s}) - \frac{1}{2}(9.80 \text{ m/s}^2)(3.02 \text{ s})^2 = 44.7 \text{ m}\end{aligned}$$

(c) We'll find the horizontal range in two steps. First, we find the time t_2 when $y = 0$ (the ball is at ground level):

$$y = 0 = v_{0y}t_2 - \frac{1}{2}gt_2^2 = t_2(v_{0y} - \frac{1}{2}gt_2)$$

Figure 3.23 Our sketch for this problem.



This is a quadratic equation for t_2 . It has two roots:

$$t_2 = 0 \quad \text{and} \quad t_2 = \frac{2v_{0y}}{g} = \frac{2(29.6 \text{ m/s})}{9.80 \text{ m/s}^2} = 6.04 \text{ s}$$

The ball is at $y = 0$ at both times. The ball *leaves* the ground at $t_2 = 0$, and it hits the ground at $t_2 = 2v_{0y}/g = 6.04$ s.

The horizontal range R is the value of x when the ball returns to the ground at $t_2 = 6.04$ s:

$$R = v_{0x}t_2 = (22.2 \text{ m/s})(6.04 \text{ s}) = 134 \text{ m}$$

The vertical component of velocity when the ball hits the ground is

$$v_y = v_{0y} - gt_2 = 29.6 \text{ m/s} - (9.80 \text{ m/s}^2)(6.04 \text{ s}) = -29.6 \text{ m/s}$$

That is, v_y has the same magnitude as the initial vertical velocity v_{0y} but the opposite direction (down). Since v_x is constant, the angle $\alpha = -53.1^\circ$ (below the horizontal) at this point is the negative of the initial angle $\alpha_0 = 53.1^\circ$.

EVALUATE As a check on our results, we can also find the maximum height in part (b) by applying the constant-acceleration formula Eq. (2.13) to the y -motion:

$$v_y^2 = v_{0y}^2 + 2a_y(y - y_0) = v_{0y}^2 - 2g(y - y_0)$$

At the highest point, $v_y = 0$ and $y = h$. Solve this equation for h ; you should get the answer that we obtained in part (b). (Do you?)

Note that the time to hit the ground, $t_2 = 6.04$ s, is exactly twice the time to reach the highest point, $t_1 = 3.02$ s. Hence the time of descent equals the time of ascent. This is *always* true if the starting point and end point are at the same elevation and if air resistance can be ignored.

Note also that $h = 44.7$ m in part (b) is comparable to the 61.0 m height above second base of the roof at Marlins Park in Miami, and the horizontal range $R = 134$ m in part (c) is greater than the 99.7 m distance from home plate to the right-field fence at Safeco Field in Seattle. In reality, due to air resistance (which we have ignored) a batted ball with the initial speed and angle we've used here won't go as high or as far as we've calculated (see Fig. 3.20).

KEYCONCEPT You can solve most projectile problems by using the equations for x , y , v_x , and v_y as functions of time. The highest point of a projectile's motion occurs at the time its vertical component of velocity is zero.

EXAMPLE 3.8 Height and range of a projectile II: Maximum height, maximum range**WITH VARIATION PROBLEMS**

Find the maximum height h and horizontal range R (see Fig. 3.23) of a projectile launched with speed v_0 at an initial angle α_0 between 0 and 90° . For a given v_0 , what value of α_0 gives maximum height? What value gives maximum horizontal range?

IDENTIFY and SET UP This is almost the same as parts (b) and (c) of Example 3.7, except that now we want general expressions for h and R . We also want the values of α_0 that give the maximum values of h and R . In part (b) of Example 3.7 we found that the projectile reaches the high point of its trajectory (so that $v_y = 0$) at time $t_1 = v_{0y}/g$, and in part (c) we found that the projectile returns to its starting height (so that $y = y_0$) at time $t_2 = 2v_{0y}/g = 2t_1$. We'll use Eq. (3.20) to find the y -coordinate h at t_1 and Eq. (3.19) to find the x -coordinate R at time t_2 . We'll express our answers in terms of the launch speed v_0 and launch angle α_0 by using Eqs. (3.18).

EXECUTE From Eqs. (3.18), $v_{0x} = v_0 \cos \alpha_0$ and $v_{0y} = v_0 \sin \alpha_0$. Hence we can write the time t_1 when $v_y = 0$ as

$$t_1 = \frac{v_{0y}}{g} = \frac{v_0 \sin \alpha_0}{g}$$

Equation (3.20) gives the height $y = h$ at this time:

$$h = (v_0 \sin \alpha_0) \left(\frac{v_0 \sin \alpha_0}{g} \right) - \frac{1}{2} g \left(\frac{v_0 \sin \alpha_0}{g} \right)^2 = \frac{v_0^2 \sin^2 \alpha_0}{2g}$$

For a given launch speed v_0 , the maximum value of h occurs for $\sin \alpha_0 = 1$ and $\alpha_0 = 90^\circ$ —that is, when the projectile is launched straight up. (If it is launched horizontally, as in Example 3.6, $\alpha_0 = 0$ and the maximum height is zero!)

The time t_2 when the projectile hits the ground is

$$t_2 = \frac{2v_{0y}}{g} = \frac{2v_0 \sin \alpha_0}{g}$$

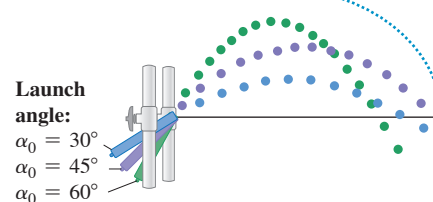
The horizontal range R is the value of x at this time. From Eq. (3.19), this is

$$R = (v_0 \cos \alpha_0) t_2 = (v_0 \cos \alpha_0) \frac{2v_0 \sin \alpha_0}{g} = \frac{v_0^2 \sin 2\alpha_0}{g}$$

(We used the trigonometric identity $2 \sin \alpha_0 \cos \alpha_0 = \sin 2\alpha_0$, found in Appendix B.) The maximum value of $\sin 2\alpha_0$ is 1; this occurs when

Figure 3.24 A launch angle of 45° gives the maximum horizontal range. The range is shorter with launch angles of 30° and 60° .

A 45° launch angle gives the greatest range; other angles fall shorter.



$2\alpha_0 = 90^\circ$, or $\alpha_0 = 45^\circ$. This angle gives the maximum range for a given initial speed if air resistance can be ignored.

EVALUATE Figure 3.24 is based on a composite photograph of three trajectories of a ball projected from a small spring gun at angles of 30° , 45° , and 60° . The initial speed v_0 is approximately the same in all three cases. The horizontal range is greatest for the 45° angle. The ranges are nearly the same for the 30° and 60° angles: Can you prove that for a given value of v_0 the range is the same for both an initial angle α_0 and an initial angle $90^\circ - \alpha_0$? (This is not the case in Fig. 3.24 due to air resistance.)

CAUTION **Height and range of a projectile** We don't recommend memorizing the above expressions for h and R . They are applicable only in the special circumstances we've described. In particular, you can use the expression for the range R *only* when launch and landing heights are equal. There are many end-of-chapter problems to which these equations do *not* apply.

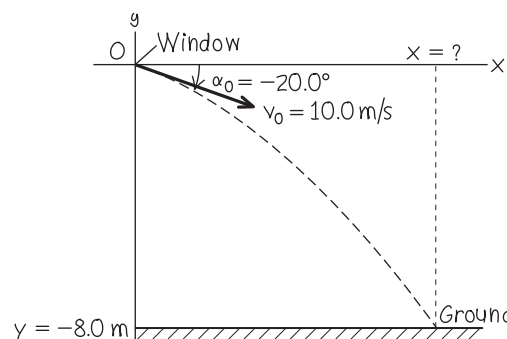
KEYCONCEPT When you solve physics problems in general, and projectile problems in particular, it's best to use symbols rather than numbers as far into the solution as possible. This allows you to better explore and understand your result.

EXAMPLE 3.9 Different initial and final heights**WITH VARIATION PROBLEMS**

You throw a ball from your window 8.0 m above the ground. When the ball leaves your hand, it is moving at 10.0 m/s at an angle of 20.0° below the horizontal. How far horizontally from your window will the ball hit the ground? Ignore air resistance.

IDENTIFY and SET UP As in Examples 3.7 and 3.8, we want to find the horizontal coordinate of a projectile when it is at a given y -value. The difference here is that this value of y is *not* the same as the initial value. We again choose the x -axis to be horizontal and the y -axis to be upward, and place the origin of coordinates at the point where the ball leaves your hand (Fig. 3.25). We have $v_0 = 10.0$ m/s and $\alpha_0 = -20.0^\circ$ (the angle is negative because the initial velocity is below the horizontal). Our target variable is the value of x when the ball reaches the ground at $y = -8.0$ m. We'll use Eq. (3.20) to find the time t when this happens and then use Eq. (3.19) to find the value of x at this time.

Figure 3.25 Our sketch for this problem.



Continued

EXECUTE To determine t , we rewrite Eq. (3.20) in the standard form for a quadratic equation for t :

$$\frac{1}{2}gt^2 - (v_0 \sin \alpha_0)t + y = 0$$

The roots of this equation are

$$\begin{aligned} t &= \frac{v_0 \sin \alpha_0 \pm \sqrt{(-v_0 \sin \alpha_0)^2 - 4\left(\frac{1}{2}g\right)y}}{2\left(\frac{1}{2}g\right)} \\ &= \frac{v_0 \sin \alpha_0 \pm \sqrt{v_0^2 \sin^2 \alpha_0 - 2gy}}{g} \\ &= \frac{\left[(10.0 \text{ m/s}) \sin(-20.0^\circ) \pm \sqrt{(10.0 \text{ m/s})^2 \sin^2(-20.0^\circ) - 2(9.80 \text{ m/s}^2)(-8.0 \text{ m})} \right]}{9.80 \text{ m/s}^2} \\ &= -1.7 \text{ s} \quad \text{or} \quad 0.98 \text{ s} \end{aligned}$$

We discard the negative root, since it refers to a time before the ball left your hand. The positive root tells us that the ball reaches the ground at $t = 0.98 \text{ s}$. From Eq. (3.19), the ball's x -coordinate at that time is

$$x = (v_0 \cos \alpha_0)t = (10.0 \text{ m/s})[\cos(-20^\circ)](0.98 \text{ s}) = 9.2 \text{ m}$$

The ball hits the ground a horizontal distance of 9.2 m from your window.

EVALUATE The root $t = -1.7 \text{ s}$ is an example of a “fictional” solution to a quadratic equation. We discussed these in Example 2.8 in Section 2.5; review that discussion.

KEYCONCEPT A projectile's vertical coordinate y as a function of time is given by a quadratic equation, which in general has more than one solution. Take care to select the solution that's appropriate for the problem you're solving.

EXAMPLE 3.10 The zookeeper and the monkey

A monkey escapes from the zoo and climbs a tree. After failing to entice the monkey down, the zookeeper fires a tranquilizer dart directly at the monkey (**Fig. 3.26**). The monkey lets go at the instant the dart leaves the gun. Show that the dart will *always* hit the monkey, provided that the dart reaches the monkey before he hits the ground and runs away.

IDENTIFY and SET UP We have *two* objects in projectile motion: the dart and the monkey. They have different initial positions and initial velocities, but they go into projectile motion at the same time $t = 0$. We'll first use Eq. (3.19) to find an expression for the time t when the x -coordinates x_{monkey} and x_{dart} are equal. Then we'll use that expression in Eq. (3.20) to see whether y_{monkey} and y_{dart} are also equal at this time; if they are, the dart hits the monkey. We make the usual choice for the x - and y -directions, and place the origin of coordinates at the muzzle of the tranquilizer gun (**Fig. 3.26**).

EXECUTE The monkey drops straight down, so $x_{\text{monkey}} = d$ at all times. From Eq. (3.19), $x_{\text{dart}} = (v_0 \cos \alpha_0)t$. We solve for the time t when these x -coordinates are equal:

$$d = (v_0 \cos \alpha_0)t \quad \text{so} \quad t = \frac{d}{v_0 \cos \alpha_0}$$

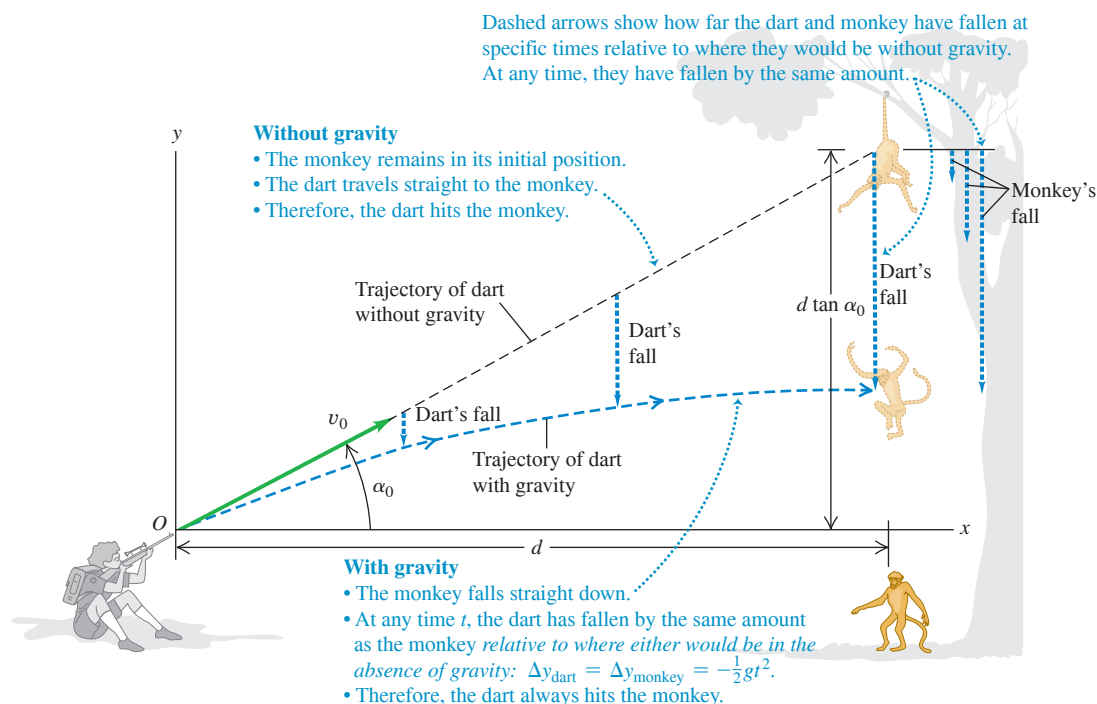
We must now show that $y_{\text{monkey}} = y_{\text{dart}}$ at this time. The monkey is in one-dimensional free fall; its position at any time is given by Eq. (2.12), with appropriate symbol changes. Figure 3.26 shows that the monkey's initial height above the dart-gun's muzzle is $y_{\text{monkey-0}} = d \tan \alpha_0$, so

$$y_{\text{monkey}} = d \tan \alpha_0 - \frac{1}{2}gt^2$$

From Eq. (3.20),

$$y_{\text{dart}} = (v_0 \sin \alpha_0)t - \frac{1}{2}gt^2$$

Figure 3.26 The tranquilizer dart hits the falling monkey.



Comparing these two equations, we see that we'll have $y_{\text{monkey}} = y_{\text{dart}}$ (and a hit) if $d \tan \alpha_0 = (v_0 \sin \alpha_0)t$ when the two x -coordinates are equal. To show that this happens, we replace t with $d/(v_0 \cos \alpha_0)$, the time when $x_{\text{monkey}} = x_{\text{dart}}$. Sure enough,

$$(v_0 \sin \alpha_0)t = (v_0 \sin \alpha_0) \frac{d}{v_0 \cos \alpha_0} = d \tan \alpha_0$$

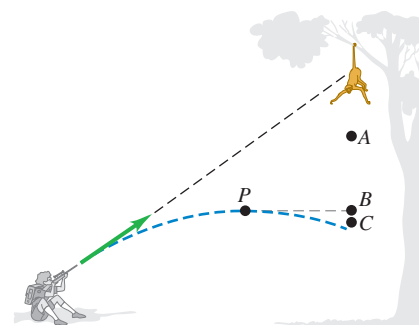
EVALUATE We've proved that the y -coordinates of the dart and the monkey are equal at the same time that their x -coordinates are equal; a dart aimed at the monkey *always* hits it, no matter what v_0

is (provided the monkey doesn't hit the ground first). This result is independent of the value of g , the acceleration due to gravity. With no gravity ($g = 0$), the monkey would remain motionless, and the dart would travel in a straight line to hit him. With gravity, both fall the same distance $\frac{1}{2}gt^2$ below their $t = 0$ positions, and the dart still hits the monkey (Fig. 3.26).

KEYCONCEPT It can be useful to think of a projectile as following a straight-line path that's pulled downward by gravity a distance $\frac{1}{2}gt^2$ in a time t .

TEST YOUR UNDERSTANDING OF SECTION 3.3 In Example 3.10, suppose the tranquilizer dart has a relatively low muzzle velocity so that the dart reaches a maximum height at a point P before striking the monkey, as shown in the figure. When the dart is at point P , will the monkey be (i) at point A (higher than P), (ii) at point B (at the same height as P), or (iii) at point C (lower than P)? Ignore air resistance.

ANSWER (i) If there were no gravity ($g = 0$), the monkey would not fall and the dart would follow a straight-line path (shown as a dashed line). The effect of gravity is to make both the monkey and the dart fall the same distance $\frac{1}{2}gt^2$ below their $t = 0$ positions. Point A is the same distance below the monkey's initial position as point P is below the dashed straight line, so point A is where we would find the monkey at the time in question.



3.4 MOTION IN A CIRCLE

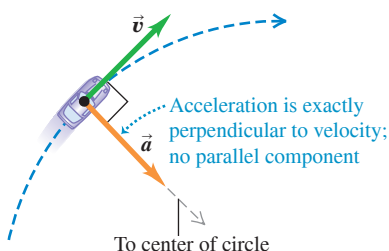
When a particle moves along a curved path, the direction of its velocity changes. As we saw in Section 3.2, this means that the particle *must* have a component of acceleration perpendicular to the path, even if its speed is constant (see Fig. 3.11b). In this section we'll calculate the acceleration for the important special case of motion in a circle.

Uniform Circular Motion

When a particle moves in a circle with *constant speed*, the motion is called **uniform circular motion**. A car rounding a curve with constant radius at constant speed, a satellite moving in a circular orbit, and an ice skater skating in a circle with constant speed are all examples of uniform circular motion (**Fig. 3.27a**; compare Fig. 3.12a). There is no component of acceleration parallel (tangent) to the path; otherwise, the speed would change. The acceleration vector is perpendicular (normal) to the path and hence directed inward (never outward!) toward the center of the circular path. This causes the direction of the velocity to change without changing the speed.

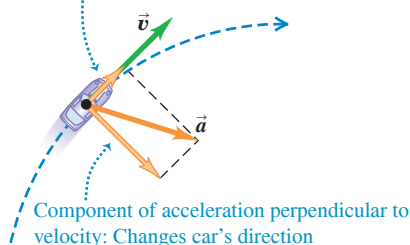
Figure 3.27 A car moving along a circular path. If the car is in uniform circular motion as in (a), the speed is constant and the acceleration is directed toward the center of the circular path (compare Fig. 3.12).

(a) **Uniform circular motion:** Constant speed along a circular path



(b) **Car speeding up along a circular path**

Component of acceleration parallel to velocity: Changes car's speed



(c) **Car slowing down along a circular path**

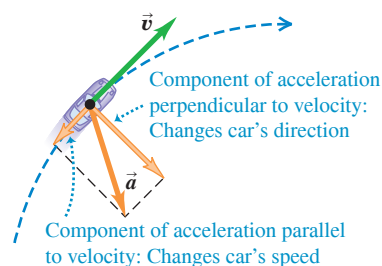
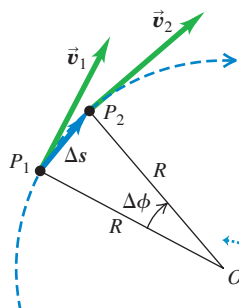
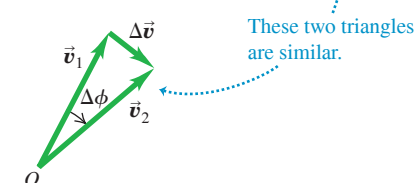


Figure 3.28 Finding the velocity change $\Delta \vec{v}$, average acceleration $\vec{a}_{\text{av}}$, and instantaneous acceleration $\vec{a}_{\text{rad}}$ for a particle moving in a circle with constant speed.

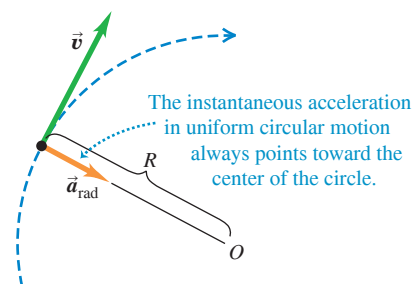
(a) A particle moves a distance Δs at constant speed along a circular path.



(b) The corresponding change in velocity $\Delta \vec{v}$. The average acceleration is in the same direction as $\Delta \vec{v}$.

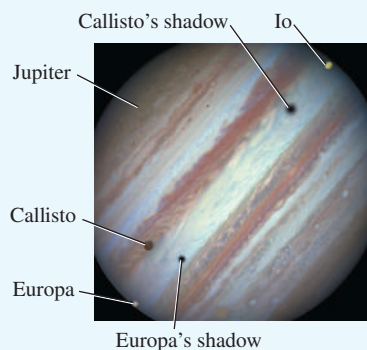


(c) The instantaneous acceleration



APPLICATION The Moons of Jupiter

Each of the three moons of Jupiter shown in this image from the Hubble Space Telescope—Io, Europa, and Callisto—moves around Jupiter in a nearly circular orbit at a nearly constant speed. The larger the radius R of a moon's orbit, the slower the speed v at which the moon moves and the smaller its centripetal acceleration $a_{\text{rad}} = v^2/R$.



We can find a simple expression for the magnitude of the acceleration in uniform circular motion. We begin with **Fig. 3.28a**, which shows a particle moving with constant speed in a circular path of radius R with center at O . The particle moves a distance Δs from P_1 to P_2 in a time interval Δt . Figure 3.28b shows the vector change in velocity $\Delta \vec{v}$ during this interval.

The angles labeled $\Delta \phi$ in Figs. 3.28a and 3.28b are the same because $\vec{v}_1$ is perpendicular to the line OP_1 and $\vec{v}_2$ is perpendicular to the line OP_2 . Hence the triangles in Figs. 3.28a and 3.28b are *similar*. The ratios of corresponding sides of similar triangles are equal, so

$$\frac{|\Delta \vec{v}|}{v_1} = \frac{\Delta s}{R} \quad \text{or} \quad |\Delta \vec{v}| = \frac{v_1}{R} \Delta s$$

The magnitude a_{av} of the average acceleration during Δt is therefore

$$a_{\text{av}} = \frac{|\Delta \vec{v}|}{\Delta t} = \frac{v_1}{R} \frac{\Delta s}{\Delta t}$$

The magnitude a of the *instantaneous* acceleration $\vec{a}$ at point P_1 is the limit of this expression as we take point P_2 closer and closer to point P_1 :

$$a = \lim_{\Delta t \rightarrow 0} \frac{v_1}{R} \frac{\Delta s}{\Delta t} = \frac{v_1}{R} \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t}$$

If the time interval Δt is short, Δs is the distance the particle moves along its curved path. So the limit of $\Delta s/\Delta t$ is the speed v_1 at point P_1 . Also, P_1 can be any point on the path, so we can drop the subscript and let v represent the speed at any point. Then

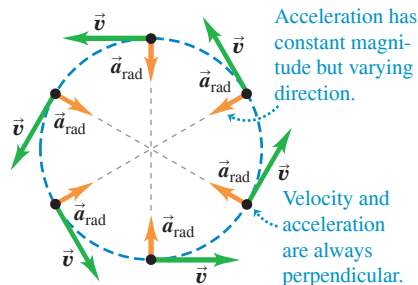
$$\text{Magnitude of acceleration of an object in uniform circular motion} \quad a_{\text{rad}} = \frac{v^2}{R} \quad \text{Speed of object} \quad \text{Radius of object's circular path} \quad (3.27)$$

The subscript “rad” is a reminder that the direction of the instantaneous acceleration at each point is always along a radius of the circle (toward the center of the circle; see Figs. 3.27a and 3.28c). So *in uniform circular motion, the magnitude a_{rad} of the instantaneous acceleration is equal to the square of the speed v divided by the radius R of the circle. Its direction is perpendicular to $\vec{v}$ and inward along the radius (Fig. 3.29a).* Because the acceleration in uniform circular motion is along the radius, we often call it **radial acceleration**.

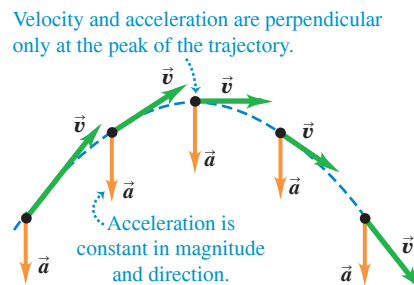
Because the acceleration in uniform circular motion is always directed toward the center of the circle, it is sometimes called **centripetal acceleration**. The word “centripetal” is derived from two Greek words meaning “seeking the center.”

Figure 3.29 Acceleration and velocity (a) for a particle in uniform circular motion and (b) for a projectile with no air resistance.

(a) Uniform circular motion



(b) Projectile motion



CAUTION Uniform circular motion vs. projectile motion Notice the differences between acceleration in uniform circular motion (Fig. 3.29a) and acceleration in projectile motion (Fig. 3.29b). It's true that in both kinds of motion the *magnitude* of acceleration is the same at all times. However, in uniform circular motion the *direction* of $\vec{a}$ changes continuously—it always points toward the center of the circle and is always perpendicular to the velocity $\vec{v}$. In projectile motion, the direction of $\vec{a}$ remains the same at all times and is perpendicular to $\vec{v}$ only at the highest point of the trajectory. 

We can also express the magnitude of the acceleration in uniform circular motion in terms of the **period** T of the motion, the time for one revolution (one complete trip around the circle). In a time T the particle travels a distance equal to the circumference $2\pi R$ of the circle, so its speed is

$$v = \frac{2\pi R}{T} \quad (3.28)$$

When we substitute this into Eq. (3.27), we obtain the alternative expression

$$\text{Magnitude of acceleration of an object in uniform circular motion} \quad a_{\text{rad}} = \frac{4\pi^2 R}{T^2} \quad \begin{array}{l} \text{Radius of object's circular path} \\ \text{Period of motion} \end{array} \quad (3.29)$$

EXAMPLE 3.11 Centripetal acceleration on a curved road

WITH  **ARIATION PROBLEMS**

An Aston Martin V12 Vantage sports car has a “lateral acceleration” of $0.97g = (0.97)(9.8 \text{ m/s}^2) = 9.5 \text{ m/s}^2$. This is the maximum centripetal acceleration the car can sustain without skidding out of a curved path. If the car is traveling at a constant 40 m/s (about 89 mi/h, or 144 km/h) on level ground, what is the radius R of the tightest unbanked curve it can negotiate?

IDENTIFY, SET UP, and EXECUTE The car is in uniform circular motion because it's moving at a constant speed along a curve that is a segment of a circle. Hence we can use Eq. (3.27) to solve for the target variable R in terms of the given centripetal acceleration a_{rad} and speed v :

$$R = \frac{v^2}{a_{\text{rad}}} = \frac{(40 \text{ m/s})^2}{9.5 \text{ m/s}^2} = 170 \text{ m (about 560 ft)}$$

This is the *minimum* turning radius because a_{rad} is the *maximum* centripetal acceleration.

EVALUATE The minimum turning radius R is proportional to the *square* of the speed, so even a small reduction in speed can make R substantially smaller. For example, reducing v by 20% (from 40 m/s to 32 m/s) would decrease R by 36% (from 170 m to 109 m).

Another way to make the minimum turning radius smaller is to *bank* the curve. We'll investigate this option in Chapter 5.

KEYCONCEPT For uniform circular motion at a given speed, decreasing the radius increases the centripetal acceleration.

EXAMPLE 3.12 Centripetal acceleration on a carnival ride

WITH  **ARIATION PROBLEMS**

Passengers on a carnival ride move at constant speed in a horizontal circle of radius 5.0 m, making a complete circle in 4.0 s. What is their acceleration?

IDENTIFY and SET UP The speed is constant, so this is uniform circular motion. We are given the radius $R = 5.0 \text{ m}$ and the period $T = 4.0 \text{ s}$, so we can use Eq. (3.29) to calculate the acceleration directly, or we can calculate the speed v by using Eq. (3.28) and then find the acceleration by using Eq. (3.27).

EXECUTE From Eq. (3.29),

$$a_{\text{rad}} = \frac{4\pi^2(5.0 \text{ m})}{(4.0 \text{ s})^2} = 12 \text{ m/s}^2 = 1.3g$$

EVALUATE We can check this answer by using the second, roundabout approach. From Eq. (3.28), the speed is

$$v = \frac{2\pi R}{T} = \frac{2\pi(5.0 \text{ m})}{4.0 \text{ s}} = 7.9 \text{ m/s}$$

The centripetal acceleration is then

$$a_{\text{rad}} = \frac{v^2}{R} = \frac{(7.9 \text{ m/s})^2}{5.0 \text{ m}} = 12 \text{ m/s}^2$$

As in Fig. 3.29a, the direction of $\vec{a}$ is always toward the center of the circle. The magnitude of $\vec{a}$ is relatively mild as carnival rides go; some roller coasters subject their passengers to accelerations as great as $4g$. To produce this acceleration with a carnival ride of radius 5.0 m would require a shorter period T and hence a faster speed v than in this example.

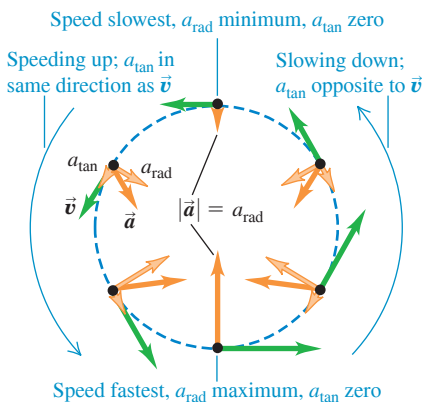
KEYCONCEPT For uniform circular motion with a given radius, decreasing the period increases the speed and the centripetal acceleration.

APPLICATION Watch Out: Tight

Curves Ahead! These roller coaster cars are in nonuniform circular motion: They slow down and speed up as they move around a vertical loop. The large accelerations involved in traveling at high speed around a tight loop mean extra stress on the passengers' circulatory systems, which is why people with cardiac conditions are cautioned against going on such rides.



Figure 3.30 A particle moving in a vertical loop with a varying speed, like a roller coaster car.

**Nonuniform Circular Motion**

We have assumed throughout this section that the particle's speed is constant as it goes around the circle. If the speed varies, we call the motion **nonuniform circular motion**. In nonuniform circular motion, Eq. (3.27) still gives the *radial* component of acceleration $a_{\text{rad}} = v^2/R$, which is always *perpendicular* to the instantaneous velocity and directed toward the center of the circle. But since the speed v has different values at different points in the motion, the value of a_{rad} is not constant. The radial (centripetal) acceleration is greatest at the point in the circle where the speed is greatest.

In nonuniform circular motion there is also a component of acceleration that is *parallel* to the instantaneous velocity (see Figs. 3.27b and 3.27c). This is the component $a_{\parallel}$ that we discussed in Section 3.2; here we call this component a_{tan} to emphasize that it is *tangent* to the circle. This component, called the **tangential acceleration** a_{tan} , is equal to the rate of change of *speed*. Thus

$$a_{\text{rad}} = \frac{v^2}{R} \quad \text{and} \quad a_{\text{tan}} = \frac{d|\vec{v}|}{dt} \quad (\text{nonuniform circular motion}) \quad (3.30)$$

The tangential component is in the same direction as the velocity if the particle is speeding up, and in the opposite direction if the particle is slowing down (**Fig. 3.30**). If the particle's speed is constant, $a_{\text{tan}} = 0$.

CAUTION Uniform vs. nonuniform circular motion The two quantities

$$\frac{d|\vec{v}|}{dt} \quad \text{and} \quad \left| \frac{d\vec{v}}{dt} \right|$$

are *not* the same. The first, equal to the tangential acceleration, is the rate of change of speed; it is zero whenever a particle moves with constant speed, even when its direction of motion changes (such as in *uniform* circular motion). The second is the magnitude of the vector acceleration; it is zero only when the particle's acceleration *vector* is zero—that is, when the particle moves in a straight line with constant speed. In *uniform* circular motion $|d\vec{v}/dt| = a_{\text{rad}} = v^2/r$; in *nonuniform* circular motion there is also a tangential component of acceleration, so $|d\vec{v}/dt| = \sqrt{a_{\text{rad}}^2 + a_{\text{tan}}^2}$.

TEST YOUR UNDERSTANDING OF SECTION 3.4 Suppose that the particle in Fig. 3.30 experiences four times the acceleration at the bottom of the loop as it does at the top of the loop. Compared to its speed at the top of the loop, is its speed at the bottom of the loop (i) $\sqrt{2}$ times as great; (ii) 2 times as great; (iii) $2\sqrt{2}$ times as great; (iv) 4 times as great; or (v) 16 times as great?

ANSWER

(ii) At both the top and bottom of the loop, the acceleration is purely radial and is given by Eq. (3.27). Radius R is the same at both points, so the difference in acceleration is due purely to differences in speed. Since a_{rad} is proportional to the square of v , the speed must be twice as great at the bottom of the loop as at the top.

3.5 RELATIVE VELOCITY

If you stand next to a one-way highway, all the cars appear to be moving forward. But if you're driving in the fast lane on that highway, slower cars appear to be moving backward. In general, when two observers measure the velocity of the same object, they get different results if one observer is moving relative to the other. The velocity seen by a particular observer is called the velocity *relative* to that observer, or simply **relative velocity**. In many situations relative velocity is extremely important (**Fig. 3.31**).

We'll first consider relative velocity along a straight line and then generalize to relative velocity in a plane.

Relative Velocity in One Dimension

A passenger walks with a velocity of $+1.0 \text{ m/s}$ along the aisle of a train that is moving with a velocity of $+3.0 \text{ m/s}$ (**Fig. 3.32a**). What is the passenger's velocity? It's a simple enough question, but it has no single answer. As seen by a second passenger sitting in the train, she is moving at $+1.0 \text{ m/s}$. A person on a bicycle standing beside the train sees the walking passenger moving at $+1.0 \text{ m/s} + 3.0 \text{ m/s} = +4.0 \text{ m/s}$. An observer in another train going in the opposite direction would give still another answer. We have to specify which observer we mean, and we speak of the velocity *relative* to a particular observer. The walking passenger's velocity relative to the train is $+1.0 \text{ m/s}$, her velocity relative to the cyclist is $+4.0 \text{ m/s}$, and so on. Each observer, equipped in principle with a meter stick and a stopwatch, forms what we call a **frame of reference**. Thus a frame of reference is a coordinate system plus a time scale.

Let's use the symbol A for the cyclist's frame of reference (at rest with respect to the ground) and the symbol B for the frame of reference of the moving train. In straight-line motion the position of a point P relative to frame A is given by $x_{P/A}$ (the position of P with respect to A), and the position of P relative to frame B is given by $x_{P/B}$ (**Fig. 3.32b**). The position of the origin of B with respect to the origin of A is $x_{B/A}$. **Figure 3.32b** shows that

$$x_{P/A} = x_{P/B} + x_{B/A} \quad (3.31)$$

In words, the coordinate of P relative to A equals the coordinate of P relative to B plus the coordinate of B relative to A .

The x -velocity of P relative to frame A , denoted by $v_{P/A-x}$, is the derivative of $x_{P/A}$ with respect to time. We can find the other velocities in the same way. So the time derivative of Eq. (3.31) gives us a relationship among the various velocities:

$$\frac{dx_{P/A}}{dt} = \frac{dx_{P/B}}{dt} + \frac{dx_{B/A}}{dt} \quad \text{or}$$

$$\text{Relative velocity along a line: } v_{P/A-x} = v_{P/B-x} + v_{B/A-x} \quad (3.32)$$

x -velocity of P relative to A x -velocity of P relative to B x -velocity of B relative to A

Getting back to the passenger on the train in **Fig. 3.32a**, we see that A is the cyclist's frame of reference, B is the frame of reference of the train, and point P represents the passenger. Using the above notation, we have

$$v_{P/B-x} = +1.0 \text{ m/s} \quad v_{B/A-x} = +3.0 \text{ m/s}$$

From Eq. (3.32) the passenger's velocity $v_{P/A-x}$ relative to the cyclist is

$$v_{P/A-x} = +1.0 \text{ m/s} + 3.0 \text{ m/s} = +4.0 \text{ m/s}$$

as we already knew.

In this example, both velocities are toward the right, and we have taken this as the positive x -direction. If the passenger walks toward the *left* relative to the train, then $v_{P/B-x} = -1.0 \text{ m/s}$, and her x -velocity relative to the cyclist is $v_{P/A-x} = -1.0 \text{ m/s} + 3.0 \text{ m/s} = +2.0 \text{ m/s}$. The sum in Eq. (3.32) is always an algebraic sum, and any or all of the x -velocities may be negative.

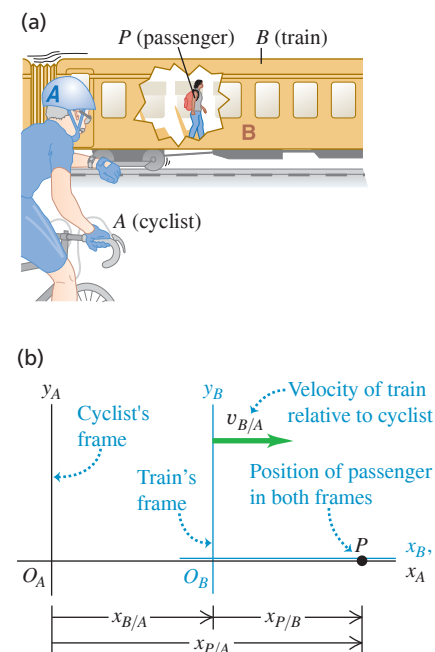
When the passenger looks out the window, the stationary cyclist on the ground appears to her to be moving backward; we call the cyclist's velocity relative to her $v_{A/P-x}$. This is just the negative of the *passenger's* velocity relative to the *cyclist*, $v_{P/A-x}$. In general, if A and B are any two points or frames of reference,

$$v_{A/B-x} = -v_{B/A-x} \quad (3.33)$$

Figure 3.31 Airshow pilots face a complicated problem involving relative velocities. They must keep track of their motion relative to the air (to maintain enough airflow over the wings to sustain lift), relative to each other (to keep a tight formation without colliding), and relative to their audience (to remain in sight of the spectators).



Figure 3.32 (a) A passenger walking in a train. (b) The position of the passenger relative to the cyclist's frame of reference and the train's frame of reference.



PROBLEM-SOLVING STRATEGY 3.2 Relative Velocity

IDENTIFY *the relevant concepts:* Whenever you see the phrase “velocity relative to” or “velocity with respect to,” it’s likely that the concepts of relative velocity will be helpful.

SET UP *the problem:* Sketch and label each frame of reference in the problem. Each moving object has its own frame of reference; in addition, you’ll almost always have to include the frame of reference of the earth’s surface. (Statements such as “The car is traveling north at 90 km/h” implicitly refer to the car’s velocity relative to the surface of the earth.) Use the labels to help identify the target variable. For example, if you want to find the x -velocity of a car (C) with respect to a bus (B), your target variable is $v_{C/B-x}$.

EXECUTE *the solution:* Solve for the target variable using Eq. (3.32). (If the velocities aren’t along the same direction, you’ll need to use the vector form of this equation, derived later in this section.) It’s

important to note the order of the double subscripts in Eq. (3.32): $v_{B/A-x}$ means “ x -velocity of B relative to A .” These subscripts obey a kind of algebra. If we regard each one as a fraction, then the fraction on the left side is the *product* of the fractions on the right side: $P/A = (P/B)(B/A)$. You can apply this rule to any number of frames of reference. For example, if there are three frames of reference A , B , and C , Eq. (3.32) becomes

$$v_{P/A-x} = v_{P/C-x} + v_{C/B-x} + v_{B/A-x}$$

EVALUATE *your answer:* Be on the lookout for stray minus signs in your answer. If the target variable is the x -velocity of a car relative to a bus ($v_{C/B-x}$), make sure that you haven’t accidentally calculated the x -velocity of the *bus* relative to the *car* ($v_{B/C-x}$). If you’ve made this mistake, you can recover by using Eq. (3.33).

EXAMPLE 3.13 Relative velocity on a straight road

WITH  **VARIATION PROBLEMS**

You drive north on a straight two-lane road at a constant 88 km/h. A truck in the other lane approaches you at a constant 104 km/h (Fig. 3.33). Find (a) the truck’s velocity relative to you and (b) your velocity relative to the truck. (c) How do the relative velocities change after you and the truck pass each other? Treat this as a one-dimensional problem.

IDENTIFY and SET UP In this problem about relative velocities along a line, there are three reference frames: you (Y), the truck (T), and the earth’s surface (E). Let the positive x -direction be north (Fig. 3.33). Then your x -velocity relative to the earth is $v_{Y/E-x} = +88$ km/h. The truck is initially approaching you, so it is moving south and its x -velocity with respect to the earth is $v_{T/E-x} = -104$ km/h. The target variables in parts (a) and (b) are $v_{T/Y-x}$ and $v_{Y/T-x}$, respectively. We’ll use Eq. (3.32) to find the first target variable and Eq. (3.33) to find the second.

EXECUTE (a) To find $v_{T/Y-x}$, we write Eq. (3.32) for the known $v_{T/E-x}$ and rearrange:

$$v_{T/E-x} = v_{T/Y-x} + v_{Y/E-x}$$

$$v_{T/Y-x} = v_{T/E-x} - v_{Y/E-x} = -104 \text{ km/h} - 88 \text{ km/h} = -192 \text{ km/h}$$

The truck is moving at 192 km/h in the negative x -direction (south) relative to you.

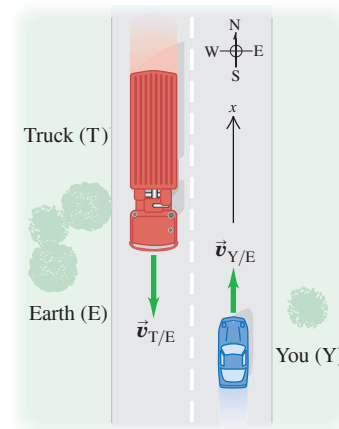
(b) From Eq. (3.33),

$$v_{Y/T-x} = -v_{T/Y-x} = -(-192 \text{ km/h}) = +192 \text{ km/h}$$

You are moving at 192 km/h in the positive x -direction (north) relative to the truck.

(c) The relative velocities do *not* change after you and the truck pass each other. The relative *positions* of the objects don’t matter. After

Figure 3.33 Reference frames for you and the truck.



it passes you, the truck is still moving at 192 km/h toward the south relative to you, even though it is now moving away from you instead of toward you.

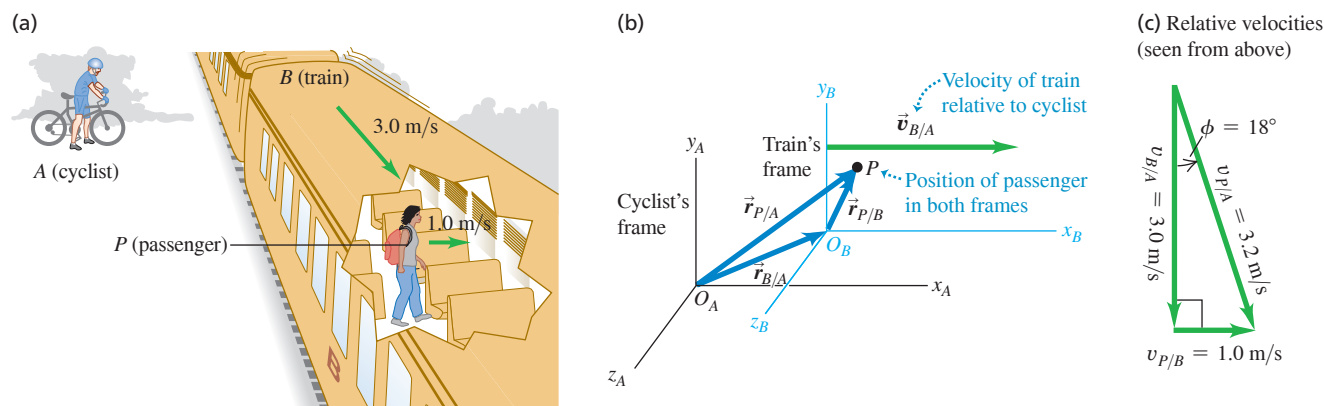
EVALUATE To check your answer in part (b), use Eq. (3.32) directly in the form $v_{Y/T-x} = v_{Y/E-x} + v_{E/T-x}$. (The x -velocity of the earth with respect to the truck is the opposite of the x -velocity of the truck with respect to the earth: $v_{E/T-x} = -v_{T/E-x}$.) Do you get the same result?

KEYCONCEPT To solve problems involving relative velocity along a line, use Eq. (3.32) and pay careful attention to the subscripts for the frames of reference in the problem.

Relative Velocity in Two or Three Dimensions

Let’s extend the concept of relative velocity to include motion in a plane or in space. Suppose that the passenger in Fig. 3.32a is walking not down the aisle of the railroad car but from one side of the car to the other, with a speed of 1.0 m/s (Fig. 3.34a). We can again describe the passenger’s position P in two frames of reference: A for the stationary ground observer and B for the moving train. But instead of coordinates x , we use position

Figure 3.34 (a) A passenger walking across a railroad car. (b) Position of the passenger relative to the cyclist's frame and the train's frame. (c) Vector diagram for the velocity of the passenger relative to the ground (the cyclist's frame), $\vec{v}_{P/A}$.



vectors $\vec{r}$ because the problem is now two-dimensional. Then, as Fig. 3.34b shows,

$$\vec{r}_{P/A} = \vec{r}_{P/B} + \vec{r}_{B/A} \quad (3.34)$$

Just as we did before, we take the time derivative of this equation to get a relationship among the various velocities; the velocity of P relative to A is $\vec{v}_{P/A} = d\vec{r}_{P/A}/dt$ and so on for the other velocities. We get

$$\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A} \quad (3.35)$$

Relative velocity in space:
Velocity of P relative to A Velocity of P relative to B Velocity of B relative to A

Equation (3.35) is known as the *Galilean velocity transformation*. It relates the velocity of an object P with respect to frame A and its velocity with respect to frame B ($\vec{v}_{P/A}$ and $\vec{v}_{P/B}$, respectively) to the velocity of frame B with respect to frame A ($\vec{v}_{B/A}$). If all three of these velocities lie along the same line, then Eq. (3.35) reduces to Eq. (3.32) for the components of the velocities along that line.

If the train is moving at $v_{B/A} = 3.0$ m/s relative to the ground and the passenger is moving at $v_{P/B} = 1.0$ m/s relative to the train, then the passenger's velocity vector $\vec{v}_{P/A}$ relative to the ground is as shown in Fig. 3.34c. The Pythagorean theorem then gives us

$$v_{P/A} = \sqrt{(3.0 \text{ m/s})^2 + (1.0 \text{ m/s})^2} = \sqrt{10 \text{ m}^2/\text{s}^2} = 3.2 \text{ m/s}$$

Figure 3.34c also shows that the *direction* of the passenger's velocity vector relative to the ground makes an angle ϕ with the train's velocity vector $\vec{v}_{B/A}$, where

$$\tan \phi = \frac{v_{P/B}}{v_{B/A}} = \frac{1.0 \text{ m/s}}{3.0 \text{ m/s}} \quad \text{and} \quad \phi = 18^\circ$$

As in the case of motion along a straight line, we have the general rule that if A and B are any two points or frames of reference,

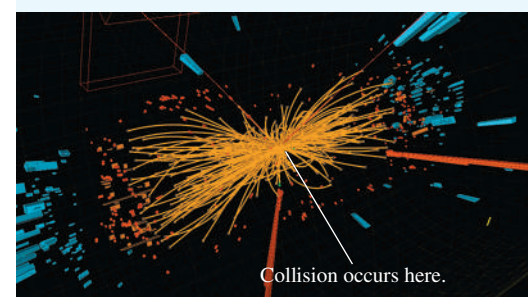
$$\vec{v}_{A/B} = -\vec{v}_{B/A} \quad (3.36)$$

The velocity of the passenger relative to the train is the negative of the velocity of the train relative to the passenger, and so on.

In the early 20th century Albert Einstein showed that Eq. (3.35) has to be modified when speeds approach the speed of light, denoted by c . It turns out that if the passenger in Fig. 3.32a could walk down the aisle at $0.30c$ and the train could move at $0.90c$, then her speed relative to the ground would be not $1.20c$ but $0.94c$; nothing can travel faster than light! We'll return to Einstein and his *special theory of relativity* in Chapter 37.

APPLICATION Relative Velocities near the Speed of Light

This image shows a spray of subatomic particles produced by the head-on collision of two protons moving in opposite directions. Relative to the laboratory, before the collision each proton is moving only 11 m/s slower than the speed of light, $c = 3.00 \times 10^8$ m/s. According to the Galilean velocity transformation, the velocity of one proton relative to the other should be approximately $c + c = 2c$. But Einstein's special theory of relativity shows that the relative velocity of the two protons must be slower than the speed of light: In fact it is equal to c minus 0.19 millimeter per second.



EXAMPLE 3.14 Flying in a crosswind**WITH VARIATION PROBLEMS**

An airplane's compass indicates that it is headed due north, and its air-speed indicator shows that it is moving through the air at 240 km/h. If there is a 100 km/h wind from west to east, what is the velocity of the airplane relative to the earth?

IDENTIFY and SET UP This problem involves velocities in two dimensions (northward and eastward), so it is a relative-velocity problem using vectors. We are given the magnitude and direction of the velocity of the plane (P) relative to the air (A). We are also given the magnitude and direction of the wind velocity, which is the velocity of the air A with respect to the earth (E):

$$\vec{v}_{P/A} = 240 \text{ km/h} \quad \text{due north}$$

$$\vec{v}_{A/E} = 100 \text{ km/h} \quad \text{due east}$$

We'll use Eq. (3.35) to find our target variables: the magnitude and direction of velocity $\vec{v}_{P/E}$ of the plane relative to the earth.

EXECUTE From Eq. (3.35) we have

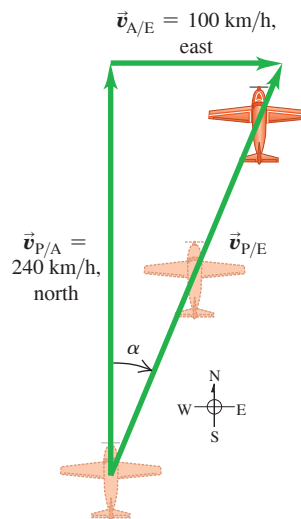
$$\vec{v}_{P/E} = \vec{v}_{P/A} + \vec{v}_{A/E}$$

Figure 3.35 shows that the three relative velocities constitute a right-triangle vector addition; the unknowns are the speed $v_{P/E}$ and the angle α . We find

$$v_{P/E} = \sqrt{(240 \text{ km/h})^2 + (100 \text{ km/h})^2} = 260 \text{ km/h}$$

$$\alpha = \arctan\left(\frac{100 \text{ km/h}}{240 \text{ km/h}}\right) = 23^\circ \text{ E of N}$$

Figure 3.35 The plane is pointed north, but the wind blows east, giving the resultant velocity $\vec{v}_{P/E}$ relative to the earth.



EVALUATE You can check the results by taking measurements on the scale drawing in Fig. 3.35. The crosswind increases the speed of the airplane relative to the earth, but pushes the airplane off course.

KEYCONCEPT To solve problems involving relative velocity in a plane or in space, use Eq. (3.35). Pay careful attention to the subscripts for the frames of reference in the problem.

EXAMPLE 3.15 Correcting for a crosswind**WITH VARIATION PROBLEMS**

With wind and airspeed as in Example 3.14, in what direction should the pilot head to travel due north? What will be her velocity relative to the earth?

IDENTIFY and SET UP Like Example 3.14, this is a relative-velocity problem with vectors. **Figure 3.36** is a scale drawing of the situation. Again the vectors add in accordance with Eq. (3.35) and form a right triangle:

$$\vec{v}_{P/E} = \vec{v}_{P/A} + \vec{v}_{A/E}$$

As Fig. 3.36 shows, the pilot points the nose of the airplane at an angle β into the wind to compensate for the crosswind. This angle, which tells us the direction of the vector $\vec{v}_{P/A}$ (the velocity of the airplane relative to the air), is one of our target variables. The other target variable is the speed of the airplane over the ground, which is the magnitude of the vector $\vec{v}_{P/E}$ (the velocity of the airplane relative to the earth). The known and unknown quantities are

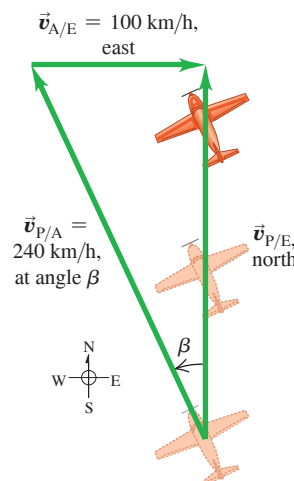
$$\vec{v}_{P/E} = \text{magnitude unknown} \quad \text{due north}$$

$$\vec{v}_{P/A} = 240 \text{ km/h} \quad \text{direction unknown}$$

$$\vec{v}_{A/E} = 100 \text{ km/h} \quad \text{due east}$$

We'll solve for the target variables by using Fig. 3.36 and trigonometry.

Figure 3.36 The pilot must point the plane in the direction of the vector $\vec{v}_{P/A}$ to travel due north relative to the earth.



EXECUTE From Fig. 3.36 the speed $v_{P/E}$ and the angle β are

$$v_{P/E} = \sqrt{(240 \text{ km/h})^2 - (100 \text{ km/h})^2} = 218 \text{ km/h}$$

$$\beta = \arcsin\left(\frac{100 \text{ km/h}}{240 \text{ km/h}}\right) = 25^\circ$$

The pilot should point the airplane 25° west of north, and her ground speed is then 218 km/h.

EVALUATE There were two target variables—the magnitude of a vector and the direction of a vector—in both this example and Example

3.14. In Example 3.14 the magnitude and direction referred to the *same* vector ($\vec{v}_{P/E}$); here they refer to *different* vectors ($\vec{v}_{P/E}$ and $\vec{v}_{P/A}$).

While we expect a *headwind* to reduce an airplane's speed relative to the ground, this example shows that a *crosswind* does, too. That's an unfortunate fact of aeronautical life.

KEYCONCEPT The vector equation for relative velocity in a plane, Eq. (3.35), allows you to solve for two unknowns, such as an unknown vector magnitude and an unknown direction.

TEST YOUR UNDERSTANDING OF SECTION 3.5 Suppose the nose of an airplane is pointed due east and the airplane has an airspeed of 150 km/h. Due to the wind, the airplane is moving due *north* relative to the ground and its speed relative to the ground is 150 km/h. What is the velocity of the air relative to the earth? (i) 150 km/h from east to west; (ii) 150 km/h from south to north; (iii) 150 km/h from southeast to northwest; (iv) 212 km/h from east to west; (v) 212 km/h from south to north; (vi) 212 km/h from southeast to northwest; (vii) there is no possible wind velocity that could cause this.

ANSWER

(vi) The effect of the wind is to cancel the airplane's eastward motion and give it a northward motion. So the velocity of the air relative to the ground (the wind velocity) must have one 150 km/h component to the west and one 150 km/h component to the north. The combination of these is a vector of magnitude $\sqrt{(150 \text{ km/h})^2 + (150 \text{ km/h})^2} = 212 \text{ km/h}$ that points to the northwest.

CHAPTER 3 SUMMARY

Position, velocity, and acceleration vectors: The position vector $\vec{r}$ of a point P in space is the vector from the origin to P . Its components are the coordinates x , y , and z .

The average velocity vector $\vec{v}_{av}$ during the time interval Δt is the displacement $\Delta\vec{r}$ (the change in position vector $\vec{r}$) divided by Δt . The instantaneous velocity vector $\vec{v}$ is the time derivative of $\vec{r}$, and its components are the time derivatives of x , y , and z . The instantaneous speed is the magnitude of $\vec{v}$. The velocity $\vec{v}$ of a particle is always tangent to the particle's path. (See Example 3.1.)

The average acceleration vector $\vec{a}_{av}$ during the time interval Δt equals $\Delta\vec{v}$ (the change in velocity vector $\vec{v}$) divided by Δt . The instantaneous acceleration vector $\vec{a}$ is the time derivative of $\vec{v}$, and its components are the time derivatives of v_x , v_y , and v_z . (See Example 3.2.)

The component of acceleration parallel to the direction of the instantaneous velocity affects the speed, while the component of $\vec{a}$ perpendicular to $\vec{v}$ affects the direction of motion. (See Examples 3.3 and 3.4.)

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \quad (3.1)$$

$$\vec{v}_{av} = \frac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1} = \frac{\Delta\vec{r}}{\Delta t} \quad (3.2)$$

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{r}}{\Delta t} = \frac{d\vec{r}}{dt} \quad (3.3)$$

$$v_x = \frac{dx}{dt} \quad v_y = \frac{dy}{dt} \quad v_z = \frac{dz}{dt} \quad (3.4)$$

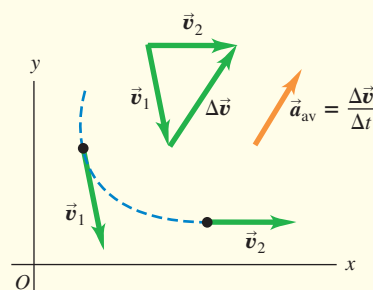
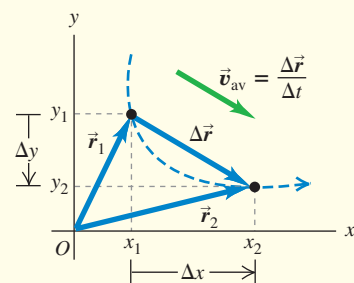
$$\vec{a}_{av} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} = \frac{\Delta\vec{v}}{\Delta t} \quad (3.8)$$

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} \quad (3.9)$$

$$a_x = \frac{dv_x}{dt} \quad (3.10)$$

$$a_y = \frac{dv_y}{dt}$$

$$a_z = \frac{dv_z}{dt}$$



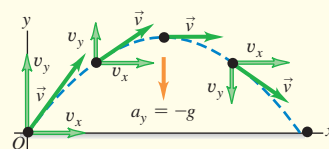
Projectile motion: In projectile motion with no air resistance, $a_x = 0$ and $a_y = -g$. The coordinates and velocity components are simple functions of time, and the shape of the path is always a parabola. We usually choose the origin to be at the initial position of the projectile. (See Examples 3.5–3.10.)

$$x = (v_0 \cos \alpha_0)t \quad (3.19)$$

$$y = (v_0 \sin \alpha_0)t - \frac{1}{2}gt^2 \quad (3.20)$$

$$v_x = v_0 \cos \alpha_0 \quad (3.21)$$

$$v_y = v_0 \sin \alpha_0 - gt \quad (3.22)$$

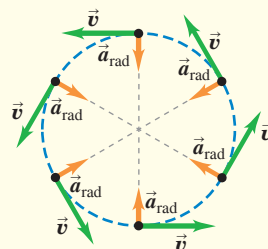


Uniform and nonuniform circular motion: When a particle moves in a circular path of radius R with constant speed v (uniform circular motion), its acceleration $\vec{a}$ is directed toward the center of the circle and perpendicular to $\vec{v}$. The magnitude a_{rad} of this radial acceleration can be expressed in terms of v and R or in terms of R and the period T (the time for one revolution), where $v = 2\pi R/T$. (See Examples 3.11 and 3.12.)

If the speed is not constant in circular motion (nonuniform circular motion), there is still a radial component of $\vec{a}$ given by Eq. (3.27) or (3.29), but there is also a component of $\vec{a}$ parallel (tangential) to the path. This tangential component is equal to the rate of change of speed, dv/dt .

$$a_{rad} = \frac{v^2}{R} \quad (3.27)$$

$$a_{rad} = \frac{4\pi^2 R}{T^2} \quad (3.29)$$



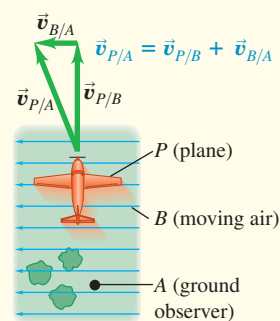
Relative velocity: When an object P moves relative to an object (or reference frame) B , and B moves relative to an object (or reference frame) A , we denote the velocity of P relative to B by $\vec{v}_{P/B}$, the velocity of P relative to A by $\vec{v}_{P/A}$, and the velocity of B relative to A by $\vec{v}_{B/A}$. If these velocities are all along the same line, their components along that line are related by Eq. (3.32). More generally, these velocities are related by Eq. (3.35). (See Examples 3.13–3.15.)

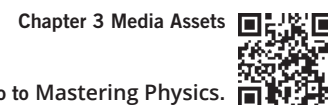
$$v_{P/A-x} = v_{P/B-x} + v_{B/A-x} \quad (3.32)$$

(relative velocity along a line)

$$\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A} \quad (3.35)$$

(relative velocity in space)





GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 3.6, 3.7, 3.8, and 3.9** (Section 3.3) before attempting these problems. In all problems, ignore air resistance.

VP3.9.1 You launch a projectile from level ground at a speed of 25.0 m/s and an angle of 36.9° above the horizontal. (a) How long after it is launched does the projectile reach its maximum height above the ground, and what is that maximum height? (b) How long after the projectile is launched does it return to ground level, and how far from its launch point does it land?

VP3.9.2 You throw a baseball at an angle of 30.0° above the horizontal. It reaches the highest point of its trajectory 1.05 s later. (a) At what speed does the baseball leave your hand? (b) What is the maximum height above the launch point that the baseball reaches?

VP3.9.3 You toss a walnut at a speed of 15.0 m/s at an angle of 50.0° above the horizontal. The launch point is on the roof of a building that is 20.0 m above the ground. (a) How long after it is launched does the walnut reach the ground? (b) How far does the walnut travel horizontally from launch point to landing point? (c) What are the horizontal and vertical components of the walnut's velocity just before it reaches the ground?

VP3.9.4 You use a slingshot to launch a potato horizontally from the edge of a cliff with speed v_0 . The acceleration due to gravity is g . Take the origin at the launch point. (a) How long after you launch the potato has it moved as far horizontally from the launch point as it has moved vertically? What are the coordinates of the potato at this time? (b) How long after you launch the potato is it moving in a direction exactly 45° below the horizontal? What are the coordinates of the potato at this time?

Be sure to review **EXAMPLES 3.11 and 3.12** (Section 3.4) before attempting these problems.

VP3.12.1 A cyclist going around a circular track at 10.0 m/s has a centripetal acceleration of 5.00 m/s^2 . What is the radius of the curve?

VP3.12.2 A race car is moving at 40.0 m/s around a circular racetrack of radius 265 m. Calculate (a) the period of the motion and (b) the car's centripetal acceleration.

VP3.12.3 The wheel of a stationary exercise bicycle at your gym makes one rotation in 0.670 s. Consider two points on this wheel: Point P is 10.0 cm from the rotation axis, and point Q is 20.0 cm from the rotation axis. Find (a) the speed of each point on the spinning wheel and (b) the centripetal acceleration of each point. (c) For points on this spinning wheel, as the distance from the axis increases, does the speed increase or decrease? Does the centripetal acceleration increase or decrease?

VP3.12.4 The planets Venus, Earth, and Mars all move in approximately circular orbits around the sun. Use the data in the table to find (a) the speed of each planet in its orbit and (b) the centripetal acceleration of each planet. (c) As the size of a planet's orbit increases, does the speed increase or decrease? Does the centripetal acceleration increase or decrease?

Planet	Orbital radius (m)	Orbital period (days)
Venus	1.08×10^{11}	225
Earth	1.50×10^{11}	365
Mars	2.28×10^{11}	687

VP3.12.5 Object A is moving at speed v in a circle of radius R . Object B is moving at speed $2v$ in a circle of radius $R/2$. (a) What is the ratio of the period of object A to the period of object B ? (b) What is the ratio of the centripetal acceleration of object A to the centripetal acceleration of object B ?

Be sure to review **EXAMPLES 3.13, 3.14, and 3.15** (Section 3.5) before attempting these problems.

VP3.15.1 A police car in a high-speed chase is traveling north on a two-lane highway at 35.0 m/s. In the southbound lane of the same highway, an SUV is moving at 18.0 m/s. Take the positive x -direction to be toward the north. Find the x -velocity of (a) the police car relative to the SUV and (b) the SUV relative to the police car.

VP3.15.2 Race cars A and B are driving on the same circular race-track at the same speed of 45.0 m/s. At a given instant car A is on the north side of the track moving eastward and car B is on the south side of the track moving westward. Find the velocity vector (magnitude and direction) of (a) car A relative to car B and (b) car B relative to car A . (c) Does the relative velocity have a component along the line connecting the two cars? Are the two cars approaching each other, moving away from each other, or neither?

VP3.15.3 Two vehicles approach an intersection: a truck moving eastbound at 16.0 m/s and an SUV moving southbound at 20.0 m/s. Find the velocity vector (magnitude and direction) of (a) the truck relative to the SUV and (b) the SUV relative to the truck.

VP3.15.4 A jet is flying due north relative to the ground. The speed of the jet relative to the ground is 155 m/s. The wind at the jet's altitude is 40.0 m/s toward the northeast (45.0° north of east). Find the speed of the jet relative to the air (its airspeed) and the direction in which the pilot of the jet must point the plane so that it travels due north relative to the ground.

BRIDGING PROBLEM Launching Up an Incline

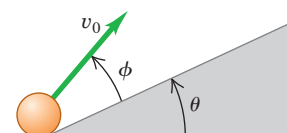
You fire a ball with an initial speed v_0 at an angle ϕ above the surface of an incline, which is itself inclined at an angle θ above the horizontal (**Fig. 3.37**). (a) Find the distance, measured along the incline, from the launch point to the point when the ball strikes the incline. (b) What angle ϕ gives the maximum range, measured along the incline? Ignore air resistance.

SOLUTION GUIDE

IDENTIFY and SET UP

1. Since there's no air resistance, this is a problem in projectile motion. The goal is to find the point where the ball's parabolic trajectory intersects the incline.

Figure 3.37 Launching a ball from an inclined ramp.



Continued

- Choose the x - and y -axes and the position of the origin. When in doubt, use the suggestions given in Problem-Solving Strategy 3.1 in Section 3.3.
- In the projectile equations in Section 3.3, the launch angle α_0 is measured from the horizontal. What is this angle in terms of θ and ϕ ? What are the initial x - and y -components of the ball's initial velocity?
- You'll need to write an equation that relates x and y for points along the incline. What is this equation? (This takes just geometry and trigonometry, not physics.)

EXECUTE

- Write the equations for the x -coordinate and y -coordinate of the ball as functions of time t .

- When the ball hits the incline, x and y are related by the equation that you found in step 4. Based on this, at what time t does the ball hit the incline?
- Based on your answer from step 6, at what coordinates x and y does the ball land on the incline? How far is this point from the launch point?
- What value of ϕ gives the *maximum* distance from the launch point to the landing point? (Use your knowledge of calculus.)

EVALUATE

- Check your answers for the case $\theta = 0$, which corresponds to the incline being horizontal rather than tilted. (You already know the answers for this case. Do you know why?)

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q3.1 A simple pendulum (a mass swinging at the end of a string) swings back and forth in a circular arc. What is the direction of the acceleration of the mass when it is at the ends of the swing? At the midpoint? In each case, explain how you obtained your answer.

Q3.2 Redraw Fig. 3.11a if $\vec{a}$ is antiparallel to $\vec{v}_1$. Does the particle move in a straight line? What happens to its speed?

Q3.3 A projectile moves in a parabolic path without air resistance. Is there any point at which $\vec{a}$ is parallel to $\vec{v}$? Perpendicular to $\vec{v}$? Explain.

Q3.4 A book slides off a horizontal tabletop. As it leaves the table's edge, the book has a horizontal velocity of magnitude v_0 . The book strikes the floor in time t . If the initial velocity of the book is doubled to $2v_0$, what happens to (a) the time the book is in the air, (b) the horizontal distance the book travels while it is in the air, and (c) the speed of the book just before it reaches the floor? In particular, does each of these quantities stay the same, double, or change in another way? Explain.

Q3.5 At the instant that you fire a bullet horizontally from a rifle, you drop a bullet from the height of the gun barrel. If there is no air resistance, which bullet hits the level ground first? Explain.

Q3.6 A package falls out of an airplane that is flying in a straight line at a constant altitude and speed. If you ignore air resistance, what would be the path of the package as observed by the pilot? As observed by a person on the ground?

Q3.7 Sketch the six graphs of the x - and y -components of position, velocity, and acceleration versus time for projectile motion with $x_0 = y_0 = 0$ and $0 < \alpha_0 < 90^\circ$.

Q3.8 If a jumping frog can give itself the same initial speed regardless of the direction in which it jumps (forward or straight up), how is the maximum vertical height to which it can jump related to its maximum horizontal range $R_{\max} = v_0^2/g$?

Q3.9 A projectile is fired upward at an angle θ above the horizontal with an initial speed v_0 . At its maximum height, what are its velocity vector, its speed, and its acceleration vector?

Q3.10 In uniform circular motion, what are the *average* velocity and *average* acceleration for one revolution? Explain.

Q3.11 In uniform circular motion, how does the acceleration change when the speed is increased by a factor of 3? When the radius is decreased by a factor of 2?

Q3.12 In uniform circular motion, the acceleration is perpendicular to the velocity at every instant. Is this true when the motion is not uniform—that is, when the speed is not constant?

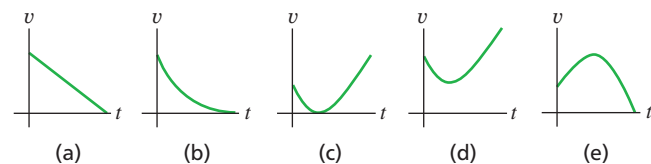
Q3.13 Raindrops hitting the side windows of a car in motion often leave diagonal streaks even if there is no wind. Why? Is the explanation the same or different for diagonal streaks on the windshield?

Q3.14 In a rainstorm with a strong wind, what determines the best position in which to hold an umbrella?

Q3.15 You are on the west bank of a river that is flowing north with a speed of 1.2 m/s. Your swimming speed relative to the water is 1.5 m/s, and the river is 60 m wide. What is your path relative to the earth that allows you to cross the river in the shortest time? Explain your reasoning.

Q3.16 A stone is thrown into the air at an angle above the horizontal and feels negligible air resistance. Which graph in Fig. Q3.16 best depicts the stone's *speed* v as a function of time t while it is in the air?

Figure Q3.16

**EXERCISES****Section 3.1 Position and Velocity Vectors**

3.1 • A squirrel has x - and y -coordinates (1.1 m, 3.4 m) at time $t_1 = 0$ and coordinates (5.3 m, -0.5 m) at time $t_2 = 3.0$ s. For this time interval, find (a) the components of the average velocity, and (b) the magnitude and direction of the average velocity.

3.2 • A rhinoceros is at the origin of coordinates at time $t_1 = 0$. For the time interval from $t_1 = 0$ to $t_2 = 12.0$ s, the rhino's average velocity has x -component -3.8 m/s and y -component 4.9 m/s. At time $t_2 = 12.0$ s, (a) what are the x - and y -coordinates of the rhino? (b) How far is the rhino from the origin?

3.3 • CALC A web page designer creates an animation in which a dot on a computer screen has position

$$\vec{r} = [4.0 \text{ cm} + (2.5 \text{ cm/s}^2)t^2]\hat{i} + (5.0 \text{ cm/s})t\hat{j}$$

(a) Find the magnitude and direction of the dot's average velocity between $t = 0$ and $t = 2.0$ s. (b) Find the magnitude and direction of the instantaneous velocity at $t = 0$, $t = 1.0$ s, and $t = 2.0$ s. (c) Sketch the dot's trajectory from $t = 0$ to $t = 2.0$ s, and show the velocities calculated in part (b).

3.4 • CALC The position of a squirrel running in a park is given by $\vec{r} = [(0.280 \text{ m/s})t + (0.0360 \text{ m/s}^2)t^2]\hat{i} + (0.0190 \text{ m/s}^3)t^3\hat{j}$. (a) What are $v_x(t)$ and $v_y(t)$, the x - and y -components of the velocity of the squirrel, as functions of time? (b) At $t = 5.00$ s, how far is the squirrel from its initial position? (c) At $t = 5.00$ s, what are the magnitude and direction of the squirrel's velocity?

Section 3.2 The Acceleration Vector

3.5 • A jet plane is flying at a constant altitude. At time $t_1 = 0$, it has components of velocity $v_x = 90 \text{ m/s}$, $v_y = 110 \text{ m/s}$. At time $t_2 = 30.0$ s, the components are $v_x = -170 \text{ m/s}$, $v_y = 40 \text{ m/s}$. (a) Sketch the velocity vectors at t_1 and t_2 . How do these two vectors differ? For this time interval calculate (b) the components of the average acceleration, and (c) the magnitude and direction of the average acceleration.

3.6 •• A dog running in an open field has components of velocity $v_x = 2.6 \text{ m/s}$ and $v_y = -1.8 \text{ m/s}$ at $t_1 = 10.0$ s. For the time interval from $t_1 = 10.0$ s to $t_2 = 20.0$ s, the average acceleration of the dog has magnitude 0.45 m/s^2 and direction 31.0° measured from the $+x$ -axis toward the $+y$ -axis. At $t_2 = 20.0$ s, (a) what are the x - and y -components of the dog's velocity? (b) What are the magnitude and direction of the dog's velocity? (c) Sketch the velocity vectors at t_1 and t_2 . How do these two vectors differ?

3.7 •• CALC The coordinates of a bird flying in the xy -plane are given by $x(t) = \alpha t$ and $y(t) = 3.0 \text{ m} - \beta t^2$, where $\alpha = 2.4 \text{ m/s}$ and $\beta = 1.2 \text{ m/s}^2$. (a) Sketch the path of the bird between $t = 0$ and $t = 2.0$ s. (b) Calculate the velocity and acceleration vectors of the bird as functions of time. (c) Calculate the magnitude and direction of the bird's velocity and acceleration at $t = 2.0$ s. (d) Sketch the velocity and acceleration vectors at $t = 2.0$ s. At this instant, is the bird's speed increasing, decreasing, or not changing? Is the bird turning? If so, in what direction?

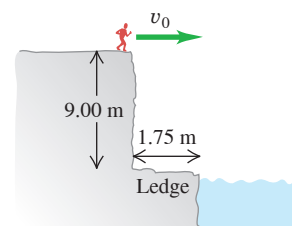
3.8 • CALC A remote-controlled car is moving in a vacant parking lot. The velocity of the car as a function of time is given by $\vec{v} = [5.00 \text{ m/s} - (0.0180 \text{ m/s}^3)t^2]\hat{i} + [2.00 \text{ m/s} + (0.550 \text{ m/s}^2)t]\hat{j}$. (a) What are $a_x(t)$ and $a_y(t)$, the x - and y -components of the car's velocity as functions of time? (b) What are the magnitude and direction of the car's velocity at $t = 8.00$ s? (c) What are the magnitude and direction of the car's acceleration at $t = 8.00$ s?

Section 3.3 Projectile Motion

3.9 • A physics book slides off a horizontal tabletop with a speed of 1.10 m/s . It strikes the floor in 0.480 s. Ignore air resistance. Find (a) the height of the tabletop above the floor; (b) the horizontal distance from the edge of the table to the point where the book strikes the floor; (c) the horizontal and vertical components of the book's velocity, and the magnitude and direction of its velocity, just before the book reaches the floor. (d) Draw x - t , y - t , v_x - t , and v_y - t graphs for the motion.

3.10 •• A daring 510 N swimmer dives off a cliff with a running horizontal leap, as shown in **Fig. E3.10**. What must her minimum speed be just as she leaves the top of the cliff so that she will miss the ledge at the bottom, which is 1.75 m wide and 9.00 m below the top of the cliff?

Figure E3.10



3.11 • Crickets Chirpy and Milada jump from the top of a vertical cliff. Chirpy drops downward and reaches the ground in 2.70 s, while Milada jumps horizontally with an initial speed of 95.0 cm/s . How far from the base of the cliff will Milada hit the ground? Ignore air resistance.

3.12 • A rookie quarterback throws a football with an initial upward velocity component of 12.0 m/s and a horizontal velocity component of 20.0 m/s . Ignore air resistance. (a) How much time is required for the football to reach the highest point of the trajectory? (b) How high is this point? (c) How much time (after it is thrown) is required for the football to return to its original level? How does this compare with the time calculated in part (a)? (d) How far has the football traveled horizontally during this time? (e) Draw x - t , y - t , v_x - t , and v_y - t graphs for the motion.

3.13 •• Leaping the River I. During a storm, a car traveling on a level horizontal road comes upon a bridge that has washed out. The driver must get to the other side, so he decides to try leaping the river with his car. The side of the road the car is on is 21.3 m above the river, while the opposite side is only 1.8 m above the river. The river itself is a raging torrent 48.0 m wide. (a) How fast should the car be traveling at the time it leaves the road in order just to clear the river and land safely on the opposite side? (b) What is the speed of the car just before it lands on the other side?

3.14 • BIO The Champion Jumper of the Insect World. The frog-hopper, *Philaenus spumarius*, holds the world record for insect jumps. When leaping at an angle of 58.0° above the horizontal, some of the tiny critters have reached a maximum height of 58.7 cm above the level ground. (See *Nature*, Vol. 424, July 31, 2003, p. 509.) Neglect air resistance in answering the following. (a) What was the takeoff speed for such a leap? (b) What horizontal distance did the froghopper cover for this world-record leap?

3.15 •• Inside a starship at rest on the earth, a ball rolls off the top of a horizontal table and lands a distance D from the foot of the table. This starship now lands on the unexplored Planet X. The commander, Captain Curious, rolls the same ball off the same table with the same initial speed as on earth and finds that it lands a distance $2.76D$ from the foot of the table. What is the acceleration due to gravity on Planet X?

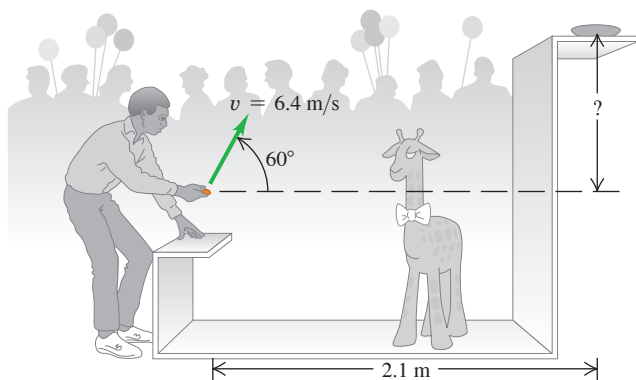
3.16 • On level ground a shell is fired with an initial velocity of 40.0 m/s at 60.0° above the horizontal and feels no appreciable air resistance. (a) Find the horizontal and vertical components of the shell's initial velocity. (b) How long does it take the shell to reach its highest point? (c) Find its maximum height above the ground. (d) How far from its firing point does the shell land? (e) At its highest point, find the horizontal and vertical components of its acceleration and velocity.

3.17 • A major leaguer hits a baseball so that it leaves the bat at a speed of 30.0 m/s and at an angle of 36.9° above the horizontal. Ignore air resistance. (a) At what two times is the baseball at a height of 10.0 m above the point at which it left the bat? (b) Calculate the horizontal and vertical components of the baseball's velocity at each of the two times calculated in part (a). (c) What are the magnitude and direction of the baseball's velocity when it returns to the level at which it left the bat?

3.18 • A shot putter releases the shot some distance above the level ground with a velocity of 12.0 m/s, 51.0° above the horizontal. The shot hits the ground 2.08 s later. Ignore air resistance. (a) What are the components of the shot's acceleration while in flight? (b) What are the components of the shot's velocity at the beginning and at the end of its trajectory? (c) How far did she throw the shot horizontally? (d) Why does the expression for R in Example 3.8 *not* give the correct answer for part (c)? (e) How high was the shot above the ground when she released it? (f) Draw x - t , y - t , v_x - t , and v_y - t graphs for the motion.

3.19 •• Win the Prize. In a carnival booth, you can win a stuffed giraffe if you toss a quarter into a small dish. The dish is on a shelf above the point where the quarter leaves your hand and is a horizontal distance of 2.1 m from this point (Fig. E3.19). If you toss the coin with a velocity of 6.4 m/s at an angle of 60° above the horizontal, the coin will land in the dish. Ignore air resistance. (a) What is the height of the shelf above the point where the quarter leaves your hand? (b) What is the vertical component of the velocity of the quarter just before it lands in the dish?

Figure E3.19



3.20 • Firefighters use a high-pressure hose to shoot a stream of water at a burning building. The water has a speed of 25.0 m/s as it leaves the end of the hose and then exhibits projectile motion. The firefighters adjust the angle of elevation α of the hose until the water takes 3.00 s to reach a building 45.0 m away. Ignore air resistance; assume that the end of the hose is at ground level. (a) Find α . (b) Find the speed and acceleration of the water at the highest point in its trajectory. (c) How high above the ground does the water strike the building, and how fast is it moving just before it hits the building?

3.21 •• A man stands on the roof of a 15.0-m-tall building and throws a rock with a speed of 30.0 m/s at an angle of 33.0° above the horizontal. Ignore air resistance. Calculate (a) the maximum height above the roof that the rock reaches; (b) the speed of the rock just before it strikes the ground; and (c) the horizontal range from the base of the building to the point where the rock strikes the ground. (d) Draw x - t , y - t , v_x - t , and v_y - t graphs for the motion.

3.22 •• At $t = 0$ a rock is projected from ground level with a speed of 15.0 m/s and at an angle of 53.0° above the horizontal. Neglect air resistance. At what two times t is the rock 5.00 m above the ground? At each of these two times, what are the horizontal and vertical components of the velocity of the rock? Let v_{0x} and v_{0y} be in the positive x - and y -directions, respectively.

3.23 • Estimate the maximum horizontal distance that you can throw a baseball. (a) Based on your estimate, what is the speed of the baseball as it leaves your hand? (b) If you could throw the baseball straight up

at the same speed as in part (a), how high would it go? (c) The value of g on Mars is 3.7 m/s^2 . What horizontal distance can you throw a baseball on Mars if you throw it with the same initial speed as in part (a)?

Section 3.4 Motion in a Circle

3.24 • Merry-go-rounds are a common ride in park playgrounds. The ride is a horizontal disk that rotates about a vertical axis at their center. A typical size is a diameter of 12 ft. A rider sits at the outer edge of the disk and holds onto a metal bar while someone pushes on the ride to make it rotate. Estimate a typical time for one rotation. (a) For your estimated time, what is the speed of the rider, in m/s? (b) What is the rider's radial acceleration, in m/s^2 ? (c) What is the rider's radial acceleration if the time for one rotation is halved?

3.25 • The earth has a radius of 6380 km and turns around once on its axis in 24 h. (a) What is the radial acceleration of an object at the earth's equator? Give your answer in m/s^2 and as a fraction of g . (b) If a_{rad} at the equator is greater than g , objects will fly off the earth's surface and into space. (We'll see the reason for this in Chapter 5.) What would the period of the earth's rotation have to be for this to occur?

3.26 •• BIO Dizziness. Our balance is maintained, at least in part, by the endolymph fluid in the inner ear. Spinning displaces this fluid, causing dizziness. Suppose that a skater is spinning very fast at 3.0 revolutions per second about a vertical axis through the center of his head. Take the inner ear to be approximately 7.0 cm from the axis of spin. (The distance varies from person to person.) What is the radial acceleration (in m/s^2 and in g 's) of the endolymph fluid?

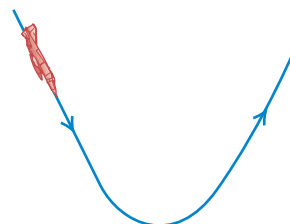
3.27 • A small ball is attached to the lower end of a 0.800-m-long string, and the other end of the string is tied to a horizontal rod. The string makes a constant angle of 37.0° with the vertical as the ball moves at a constant speed in a horizontal circle. If it takes the ball 0.600 s to complete one revolution, what is the magnitude of the radial acceleration of the ball?

3.28 •• A model of a helicopter rotor has four blades, each 3.40 m long from the central shaft to the blade tip. The model is rotated in a wind tunnel at 550 rev/min. (a) What is the linear speed of the blade tip, in m/s? (b) What is the radial acceleration of the blade tip expressed as a multiple of g ?

3.29 BIO Pilot Blackout in a Power Dive. A jet plane comes in for a downward dive as shown in Fig. E3.29. The bottom part of the path is a quarter circle with a radius of curvature of 280 m. According to medical tests, pilots will lose consciousness when they pull out of a dive at an upward acceleration greater than $5.5g$. At what minimum speed (in m/s and in mph) will the pilot black out during this dive?

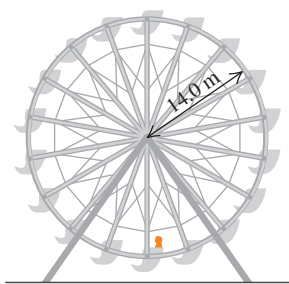
3.30 • An object moves in a horizontal circle at constant speed v (in units of m/s). It takes the object T seconds to complete one revolution. Derive an expression that gives the radial acceleration of the ball in terms of v and T , but not r . (a) If the speed doubles, by what factor must the period T change if a_{rad} is to remain unchanged? (b) If the radius doubles, by what factor must the period change to keep a_{rad} the same?

Figure E3.29



3.31 • A Ferris wheel with radius 14.0 m is turning about a horizontal axis through its center (**Fig. E3.31**). The linear speed of a passenger on the rim is constant and equal to 6.00 m/s. What are the magnitude and direction of the passenger's acceleration as she passes through (a) the lowest point in her circular motion and (b) the highest point in her circular motion? (c) How much time does it take the Ferris wheel to make one revolution?

Figure E3.31



3.32 • A roller coaster car moves in a vertical circle of radius R . At the top of the circle the car has speed v_1 , and at the bottom of the circle it has speed v_2 , where $v_2 > v_1$. (a) When the car is at the top of its circular path, what is the direction of its radial acceleration, $a_{\text{rad, top}}$? (b) When the car is at the bottom of its circular path, what is the direction of its radial acceleration, $a_{\text{rad, bottom}}$? (c) In terms of v_1 and v_2 , what is the ratio $a_{\text{rad, bottom}}/a_{\text{rad, top}}$?

3.33 •• BIO Hypergravity. At its Ames Research Center, NASA uses its large “20 G” centrifuge to test the effects of very large accelerations (“hypergravity”) on test pilots and astronauts. In this device, an arm 8.84 m long rotates about one end in a horizontal plane, and an astronaut is strapped in at the other end. Suppose that he is aligned along the centrifuge's arm with his head at the outermost end. The maximum sustained acceleration to which humans are subjected in this device is typically 12.5g. (a) How fast must the astronaut's head be moving to experience this maximum acceleration? (b) What is the difference between the acceleration of his head and feet if the astronaut is 2.00 m tall? (c) How fast in rpm (rev/min) is the arm turning to produce the maximum sustained acceleration?

3.34 • The radius of the earth's orbit around the sun (assumed to be circular) is 1.50×10^8 km, and the earth travels around this orbit in 365 days. (a) What is the magnitude of the orbital velocity of the earth, in m/s? (b) What is the magnitude of the radial acceleration of the earth toward the sun, in m/s^2 ? (c) Repeat parts (a) and (b) for the motion of the planet Mercury (orbit radius = 5.79×10^7 km, orbital period = 88.0 days).

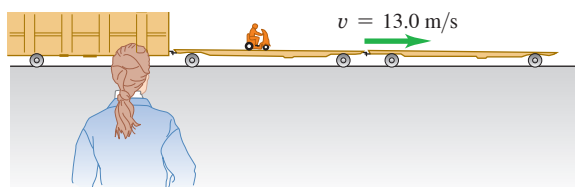
Section 3.5 Relative Velocity

3.35 • A “moving sidewalk” in an airport terminal moves at 1.0 m/s and is 35.0 m long. If a woman steps on at one end and walks at 1.5 m/s relative to the moving sidewalk, how much time does it take her to reach the opposite end if she walks (a) in the same direction the sidewalk is moving? (b) In the opposite direction?

3.36 • A railroad flatcar is traveling to the right at a speed of 13.0 m/s relative to an observer standing on the ground. Someone is riding a motor scooter on the flatcar (**Fig. E3.36**). What is the velocity (magnitude and direction) of the scooter relative to the flatcar if the scooter's velocity relative to the observer on the ground is (a) 18.0 m/s to the right? (b) 3.0 m/s to the left? (c) zero?

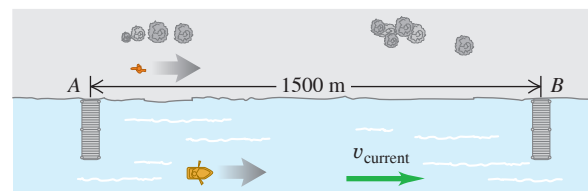
3.37 •• A canoe has a velocity of 0.40 m/s southeast relative to the earth. The canoe is on a river that is flowing 0.50 m/s east relative to the earth. Find the velocity (magnitude and direction) of the canoe relative to the river.

Figure E3.36



3.38 • Two piers, A and B, are located on a river; B is 1500 m downstream from A (**Fig. E3.38**). Two friends must make round trips from pier A to pier B and return. One rows a boat at a constant speed of 4.00 km/h relative to the water; the other walks on the shore at a constant speed of 4.00 km/h. The velocity of the river is 2.80 km/h in the direction from A to B. How much time does it take each person to make the round trip?

Figure E3.38



3.39 •• BIO Bird Migration. Canada geese migrate essentially along a north–south direction for well over a thousand kilometers in some cases, traveling at speeds up to about 100 km/h. If one goose is flying at 100 km/h relative to the air but a 40 km/h wind is blowing from west to east, (a) at what angle relative to the north–south direction should this bird head to travel directly southward relative to the ground? (b) How long will it take the goose to cover a ground distance of 500 km from north to south? (*Note:* Even on cloudy nights, many birds can navigate by using the earth's magnetic field to fix the north–south direction.)

3.40 •• The nose of an ultralight plane is pointed due south, and its airspeed indicator shows 35 m/s. The plane is in a 10 m/s wind blowing toward the southwest relative to the earth. (a) In a vector-addition diagram, show the relationship of $\vec{v}_{P/E}$ (the velocity of the plane relative to the earth) to the two given vectors. (b) Let x be east and y be north, and find the components of $\vec{v}_{P/E}$. (c) Find the magnitude and direction of $\vec{v}_{P/E}$.

3.41 • Crossing the River I. A river flows due south with a speed of 2.0 m/s. You steer a motorboat across the river; your velocity relative to the water is 4.2 m/s due east. The river is 500 m wide. (a) What is your velocity (magnitude and direction) relative to the earth? (b) How much time is required to cross the river? (c) How far south of your starting point will you reach the opposite bank?

3.42 • Crossing the River II. (a) In which direction should the motorboat in Exercise 3.41 head to reach a point on the opposite bank directly east from your starting point? (The boat's speed relative to the water remains 4.2 m/s.) (b) What is the velocity of the boat relative to the earth? (c) How much time is required to cross the river?

3.43 •• An airplane pilot wishes to fly due west. A wind of 80.0 km/h (about 50 mi/h) is blowing toward the south. (a) If the airspeed of the plane (its speed in still air) is 320.0 km/h (about 200 mi/h), in which direction should the pilot head? (b) What is the speed of the plane over the ground? Draw a vector diagram.

PROBLEMS

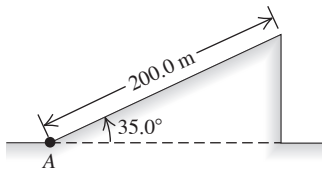
3.44 •• CPCALC A dog in an open field is at rest under a tree at time $t = 0$ and then runs with acceleration $\vec{a}(t) = (0.400 \text{ m/s}^2)\hat{i} - (0.180 \text{ m/s}^3)t\hat{j}$. How far is the dog from the tree 8.00 s after it starts to run?

3.45 •• CALC If $\vec{r} = bt^2\hat{i} + ct^3\hat{j}$, where b and c are positive constants, when does the velocity vector make an angle of 45.0° with the x - and y -axes?

3.46 ••• CALC A faulty model rocket moves in the xy -plane (the positive y -direction is vertically upward). The rocket's acceleration has components $a_x(t) = \alpha t^2$ and $a_y(t) = \beta - \gamma t$, where $\alpha = 2.50 \text{ m/s}^4$, $\beta = 9.00 \text{ m/s}^2$, and $\gamma = 1.40 \text{ m/s}^3$. At $t = 0$ the rocket is at the origin and has velocity $\vec{v}_0 = v_{0x}\hat{i} + v_{0y}\hat{j}$ with $v_{0x} = 1.00 \text{ m/s}$ and $v_{0y} = 7.00 \text{ m/s}$. (a) Calculate the velocity and position vectors as functions of time. (b) What is the maximum height reached by the rocket? (c) What is the horizontal displacement of the rocket when it returns to $y = 0$?

3.47 ••• CP A test rocket starting from rest at point A is launched by accelerating it along a 200.0 m incline at 1.90 m/s^2 (Fig. P3.47). The incline rises at 35.0° above the horizontal, and at the instant the rocket leaves it, the engines turn off and the rocket is subject to gravity only (ignore air resistance). Find (a) the maximum height above the ground that the rocket reaches, and (b) the rocket's greatest horizontal range beyond point A.

Figure P3.47



3.48 •• CALC The position of a dragonfly that is flying parallel to the ground is given as a function of time by $\vec{r} = [2.90 \text{ m} + (0.0900 \text{ m/s}^2)t^2]\hat{i} - (0.0150 \text{ m/s}^3)t^3\hat{j}$. (a) At what value of t does the velocity vector of the dragonfly make an angle of 30.0° clockwise from the $+x$ -axis? (b) At the time calculated in part (a), what are the magnitude and direction of the dragonfly's acceleration vector?

3.49 •• In fighting forest fires, airplanes work in support of ground crews by dropping water on the fires. For practice, a pilot drops a canister of red dye, hoping to hit a target on the ground below. If the plane is flying in a horizontal path 90.0 m above the ground and has a speed of 64.0 m/s (143 mi/h), at what horizontal distance from the target should the pilot release the canister? Ignore air resistance.

3.50 •• CALC A bird flies in the xy -plane with a velocity vector given by $\vec{v} = (\alpha - \beta t^2)\hat{i} + \gamma t\hat{j}$, with $\alpha = 2.4 \text{ m/s}$, $\beta = 1.6 \text{ m/s}^3$, and $\gamma = 4.0 \text{ m/s}^2$. The positive y -direction is vertically upward. At $t = 0$ the bird is at the origin. (a) Calculate the position and acceleration vectors of the bird as functions of time. (b) What is the bird's altitude (y -coordinate) as it flies over $x = 0$ for the first time after $t = 0$?

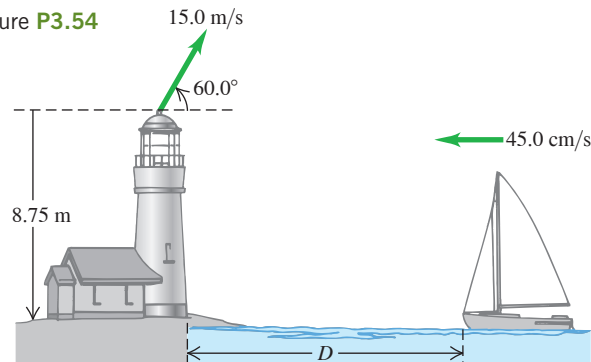
3.51 ••• A movie stuntwoman drops from a helicopter that is 30.0 m above the ground and moving with a constant velocity whose components are 10.0 m/s upward and 15.0 m/s horizontal and toward the south. Ignore air resistance. (a) Where on the ground (relative to the position of the helicopter when she drops) should the stuntwoman have placed foam mats to break her fall? (b) Draw x - t , y - t , v_x - t , and v_y - t graphs of her motion.

3.52 •• A cannon, located 60.0 m from the base of a vertical 25.0-m -tall cliff, shoots a 15 kg shell at 43.0° above the horizontal toward the cliff. (a) What must the minimum muzzle velocity be for the shell to clear the top of the cliff? (b) The ground at the top of the cliff is level, with a constant elevation of 25.0 m above the cannon. Under the conditions of part (a), how far does the shell land past the edge of the cliff?

3.53 • CP CALC A toy rocket is launched with an initial velocity of 12.0 m/s in the horizontal direction from the roof of a 30.0-m -tall building. The rocket's engine produces a horizontal acceleration of $(1.60 \text{ m/s}^3)t$, in the same direction as the initial velocity, but in the vertical direction the acceleration is g , downward. Ignore air resistance. What horizontal distance does the rocket travel before reaching the ground?

3.54 ••• An important piece of landing equipment must be thrown to a ship, which is moving at 45.0 cm/s , before the ship can dock. This equipment is thrown at 15.0 m/s at 60.0° above the horizontal from the top of a tower at the edge of the water, 8.75 m above the ship's deck (Fig. P3.54). For this equipment to land at the front of the ship, at what distance D from the dock should the ship be when the equipment is thrown? Ignore air resistance.

Figure P3.54

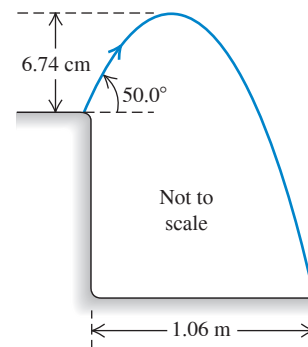


3.55 •• A baseball thrown at an angle of 60.0° above the horizontal strikes a building 18.0 m away at a point 8.00 m above the point from which it is thrown. Ignore air resistance. (a) Find the magnitude of the ball's initial velocity (the velocity with which the ball is thrown). (b) Find the magnitude and direction of the velocity of the ball just before it strikes the building.

3.56 •• An Errand of Mercy. An airplane is dropping bales of hay to cattle stranded in a blizzard on the Great Plains. The pilot releases the bales at 150 m above the level ground when the plane is flying at 75 m/s in a direction 55° above the horizontal. How far in front of the cattle should the pilot release the hay so that the bales land at the point where the cattle are stranded?

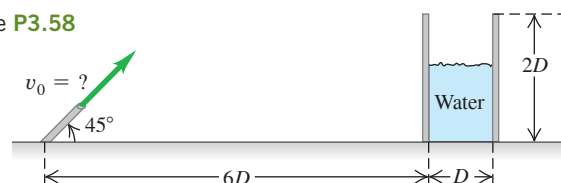
3.57 •• A grasshopper leaps into the air from the edge of a vertical cliff, as shown in Fig. P3.57. Find (a) the initial speed of the grasshopper and (b) the height of the cliff.

Figure P3.57



3.58 ••• A water hose is used to fill a large cylindrical storage tank of diameter D and height $2D$. The hose shoots the water at 45° above the horizontal from the same level as the base of the tank and is a distance $6D$ away (Fig. P3.58). For what range of launch speeds (v_0) will the water enter the tank? Ignore air resistance, and express your answer in terms of D and g .

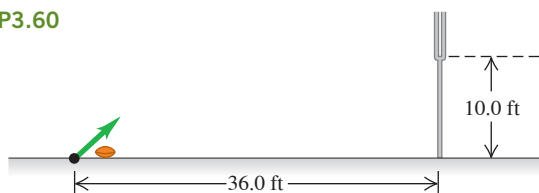
Figure P3.58



3.59 •• An object is projected with initial speed v_0 from the edge of the roof of a building that has height H . The initial velocity of the object makes an angle α_0 with the horizontal. Neglect air resistance. (a) If α_0 is 90° , so that the object is thrown straight up (but misses the roof on the way down), what is the speed v of the object just before it strikes the ground? (b) If $\alpha_0 = -90^\circ$, so that the object is thrown straight down, what is its speed just before it strikes the ground? (c) Derive an expression for the speed v of the object just before it strikes the ground for general α_0 . (d) The final speed v equals v_1 when α_0 equals α_1 . If α_0 is increased, does v increase, decrease, or stay the same?

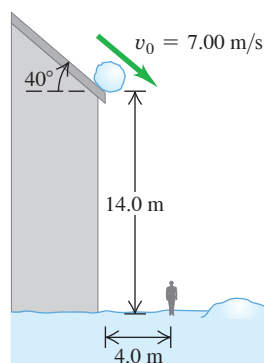
3.60 •• Kicking an Extra Point. In Canadian football, after a touchdown the team has the opportunity to earn one more point by kicking the ball over the bar between the goal posts. The bar is 10.0 ft above the ground, and the ball is kicked from ground level, 36.0 ft horizontally from the bar (**Fig. P3.60**). Football regulations are stated in English units, but convert them to SI units for this problem. (a) There is a minimum angle above the ground such that if the ball is launched below this angle, it can never clear the bar, no matter how fast it is kicked. What is this angle? (b) If the ball is kicked at 45.0° above the horizontal, what must its initial speed be if it is just to clear the bar? Express your answer in m/s and in km/h.

Figure P3.60



3.61 ••• Look Out! A snowball rolls off a barn roof that slopes downward at an angle of 40° (**Fig. P3.61**). The edge of the roof is 14.0 m above the ground, and the snowball has a speed of 7.00 m/s as it rolls off the roof. Ignore air resistance. (a) How far from the edge of the barn does the snowball strike the ground if it doesn't strike anything else while falling? (b) Draw $x-t$, $y-t$, v_x-t , and v_y-t graphs for the motion in part (a). (c) A man 1.9 m tall is standing 4.0 m from the edge of the barn. Will the snowball hit him?

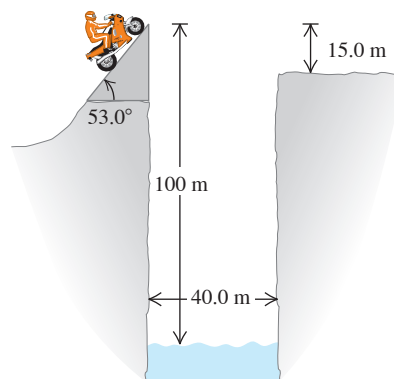
Figure P3.61



3.62 • A 2.7 kg ball is thrown upward with an initial speed of 20.0 m/s from the edge of a 45.0-m-high cliff. At the instant the ball is thrown, a woman starts running away from the base of the cliff with a constant speed of 6.00 m/s. The woman runs in a straight line on level ground. Ignore air resistance on the ball. (a) At what angle above the horizontal should the ball be thrown so that the runner will catch it just before it hits the ground, and how far does she run before she catches the ball? (b) Carefully sketch the ball's trajectory as viewed by (i) a person at rest on the ground and (ii) the runner.

3.63 •• Leaping the River II. A physics professor did daredevil stunts in his spare time. His last stunt was an attempt to jump across a river on a motorcycle (**Fig. P3.63**). The takeoff ramp was inclined at 53.0° , the river was 40.0 m wide, and the far bank was 15.0 m lower than the top of the ramp. The river itself was 100 m below the ramp. Ignore air resistance. (a) What should his speed have been at the top of the ramp to have just made it to the edge of the far bank? (b) If his speed was only half the value found in part (a), where did he land?

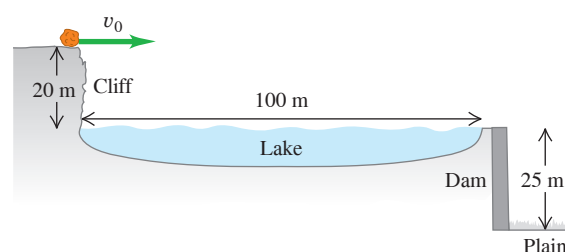
Figure P3.63



3.64 •• Tossing Your Lunch. Henrietta is jogging on the sidewalk at 3.05 m/s on the way to her physics class. Bruce realizes that she forgot her bag of bagels, so he runs to the window, which is 38.0 m above the street level and directly above the sidewalk, to throw the bag to her. He throws it horizontally 9.00 s after she has passed below the window, and she catches it on the run. Ignore air resistance. (a) With what initial speed must Bruce throw the bagels so that Henrietta can catch the bag just before it hits the ground? (b) Where is Henrietta when she catches the bagels?

3.65 • A 76.0 kg rock is rolling horizontally at the top of a vertical cliff that is 20 m above the surface of a lake (**Fig. P3.65**). The top of the vertical face of a dam is located 100 m from the foot of the cliff, with the top of the dam level with the surface of the water in the lake. A level plain is 25 m below the top of the dam. (a) What must be the minimum speed of the rock just as it leaves the cliff so that it will reach the plain without striking the dam? (b) How far from the foot of the dam does the rock hit the plain?

Figure P3.65



3.66 •• A firefighting crew uses a water cannon that shoots water at 25.0 m/s at a fixed angle of 53.0° above the horizontal. The firefighters want to direct the water at a blaze that is 10.0 m above ground level. How far from the building should they position their cannon? There are *two* possibilities; can you get them both? (*Hint:* Start with a sketch showing the trajectory of the water.)

3.67 •• On a level football field a football is projected from ground level. It has speed 8.0 m/s when it is at its maximum height. It travels a horizontal distance of 50.0 m. Neglect air resistance. How long is the ball in the air?

3.68 ••• You are standing on a loading dock at the top of a flat ramp that is at a constant angle α_0 below the horizontal. You slide a small box horizontally off the loading dock with speed v_0 in a direction so that it lands on the ramp. How far vertically downward does the box travel before it strikes the ramp?

3.69 ••• In the middle of the night you are standing a horizontal distance of 14.0 m from the high fence that surrounds the estate of your rich uncle. The top of the fence is 5.00 m above the ground. You have taped an important message to a rock that you want to throw over the fence. The ground is level, and the width of the fence is small enough to be ignored. You throw the rock from a height of 1.60 m above the ground and at an angle of 56.0° above the horizontal. (a) What minimum initial speed must the rock have as it leaves your hand to clear the top of the fence? (b) For the initial velocity calculated in part (a), what horizontal distance beyond the fence will the rock land on the ground?

3.70 ••• A small object is projected from level ground with an initial velocity of magnitude 16.0 m/s and directed at an angle of 60.0° above the horizontal. (a) What is the horizontal displacement of the object when it is at its maximum height? How does your result compare to the horizontal range R of the object? (b) What is the vertical displacement of the object when its horizontal displacement is 80.0% of its horizontal range R ? How does your result compare to the maximum height $h_{\max}$ reached by the object? (c) For when the object has horizontal displacement $x - x_0 = \alpha R$, where α is a positive constant, derive an expression (in terms of α) for $(y - y_0)/h_{\max}$. Your result should not depend on the initial velocity or the angle of projection. Show that your expression gives the correct result when $\alpha = 0.80$, as is the case in part (b). Also show that your expression gives the correct result for $\alpha = 0$, $\alpha = 0.50$, and $\alpha = 1.0$.

3.71 •• An airplane pilot sets a compass course due west and maintains an airspeed of 220 km/h. After flying for 0.500 h, she finds herself over a town 120 km west and 20 km south of her starting point. (a) Find the wind velocity (magnitude and direction). (b) If the wind velocity is 40 km/h due south, in what direction should the pilot set her course to travel due west? Use the same airspeed of 220 km/h.

3.72 •• Raindrops. When a train's velocity is 12.0 m/s eastward, raindrops that are falling vertically with respect to the earth make traces that are inclined 30.0° to the vertical on the windows of the train. (a) What is the horizontal component of a drop's velocity with respect to the earth? With respect to the train? (b) What is the magnitude of the velocity of the raindrop with respect to the earth? With respect to the train?

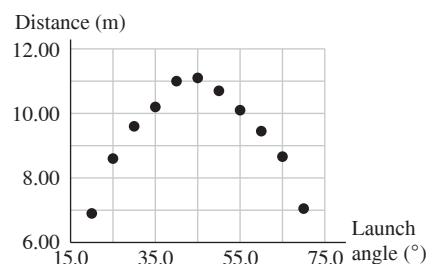
3.73 ••• In a World Cup soccer match, Juan is running due north toward the goal with a speed of 8.00 m/s relative to the ground. A teammate passes the ball to him. The ball has a speed of 12.0 m/s and is moving in a direction 37.0° east of north, relative to the ground. What are the magnitude and direction of the ball's velocity relative to Juan?

3.74 •• A shortstop is running due east as he throws a baseball to the catcher, who is standing at home plate. The velocity of the baseball relative to the shortstop is 6.00 m/s in the direction due south, and the speed of the baseball relative to the catcher is 9.00 m/s. What is the speed of the shortstop relative to the ground when he throws the ball?

3.75 •• Two soccer players, Mia and Alice, are running as Alice passes the ball to Mia. Mia is running due north with a speed of 6.00 m/s. The velocity of the ball relative to Mia is 5.00 m/s in a direction 30.0° east of south. What are the magnitude and direction of the velocity of the ball relative to the ground?

3.76 •• DATA A spring-gun projects a small rock from the ground with speed v_0 at an angle θ_0 above the ground. You have been asked to determine v_0 . From the way the spring-gun is constructed, you know that to a good approximation v_0 is independent of the launch angle. You go to a level, open field, select a launch angle, and measure the horizontal distance the rock travels. You use $g = 9.80 \text{ m/s}^2$ and ignore the small height of the end of the spring-gun's barrel above the ground. Since your measurement includes some uncertainty in values measured for the launch angle and for the horizontal range, you repeat the measurement for several launch angles and obtain the results given in **Fig. 3.76**. You ignore air resistance because there is no wind and the rock is small and heavy. (a) Select a way to represent the data well as a straight line. (b) Use the slope of the best straight-line fit to your data from part (a) to calculate v_0 . (c) When the launch angle is 36.9° , what maximum height above the ground does the rock reach?

Figure P3.76



3.77 •• DATA You have constructed a hair-spray-powered potato gun and want to find the muzzle speed v_0 of the potatoes, the speed they have as they leave the end of the gun barrel. You use the same amount of hair spray each time you fire the gun, and you have confirmed by repeated firings at the same height that the muzzle speed is approximately the same for each firing. You climb on a microwave relay tower (with permission, of course) to launch the potatoes horizontally from different heights above the ground. Your friend measures the height of the gun barrel above the ground and the range R of each potato. You obtain the data in the table.

Launch height h	Horizontal range R
2.00 m	10.4 m
6.00 m	17.1 m
9.00 m	21.3 m
12.00 m	25.8 m

Each of the values of h and R has some measurement error: The muzzle speed is not precisely the same each time, and the barrel isn't precisely horizontal. So you use all of the measurements to get the best estimate of v_0 . No wind is blowing, so you decide to ignore air resistance. You use $g = 9.80 \text{ m/s}^2$ in your analysis. (a) Select a way to represent the data well as a straight line. (b) Use the slope of the best-fit line from part (a) to calculate the average value of v_0 . (c) What would be the horizontal range of a potato that is fired from ground level at an angle of 30.0° above the horizontal? Use the value of v_0 that you calculated in part (b).

3.78 ••• DATA You are a member of a geological team in Central Africa. Your team comes upon a wide river that is flowing east. You must determine the width of the river and the current speed (the speed of the water relative to the earth). You have a small boat with an outboard motor. By measuring the time it takes to cross a pond where the water isn't flowing, you have calibrated the throttle settings to the speed of the boat in still water. You set the throttle so that the speed of the boat relative to the river is a constant 6.00 m/s. Traveling due north across the river, you reach the opposite bank in 20.1 s. For the return trip, you change the throttle setting so that the speed of the boat relative to the water is 9.00 m/s. You travel due south from one bank to the other and cross the river in 11.2 s. (a) How wide is the river, and what is the current speed? (b) With the throttle set so that the speed of the boat relative to the water is 6.00 m/s, what is the shortest time in which you could cross the river, and where on the far bank would you land?

CHALLENGE PROBLEMS

3.79 ••• CALC A projectile thrown from a point P moves in such a way that its distance from P is always increasing. Find the maximum angle above the horizontal with which the projectile could have been thrown. Ignore air resistance.

3.80 ••• Two students are canoeing on a river. While heading upstream, they accidentally drop an empty bottle overboard. They then continue paddling for 60 minutes, reaching a point 2.0 km farther upstream. At this point they realize that the bottle is missing and, driven by ecological awareness, they turn around and head downstream. They catch up with and retrieve the bottle (which has been moving along with the current) 5.0 km downstream from the turnaround point. (a) Assuming a constant paddling effort throughout, how fast is the river flowing? (b) What would the canoe speed in a still lake be for the same paddling effort?

3.81 ••• CP A rocket designed to place small payloads into orbit is carried to an altitude of 12.0 km above sea level by a converted airliner. When the airliner is flying in a straight line at a constant speed of 850 km/h, the rocket is dropped. After the drop, the airliner maintains the same altitude and speed and continues to fly in a straight line. The rocket falls for a brief time, after which its rocket motor turns on. Once that motor is on, the combined effects of thrust and gravity give the rocket a constant acceleration of magnitude $3.00g$ directed at an angle of 30.0° above the horizontal. For safety, the rocket should be at least 1.00 km in front of the airliner when it climbs through the airliner's altitude. Your job is to determine the minimum time that the rocket must fall before its engine starts. Ignore air resistance. Your answer should include (i) a diagram showing the flight paths of both the rocket and the airliner, labeled at several points with vectors for their velocities and accelerations; (ii) an x - t graph showing the motions of both the rocket and the airliner; and (iii) a y - t graph showing the motions of both the rocket and the airliner. In the diagram and the graphs, indicate when the rocket is dropped, when the rocket motor turns on, and when the rocket climbs through the altitude of the airliner.

MCAT-STYLE PASSAGE PROBLEMS

BIO Ballistic Seed Dispersal. Some plants disperse their seeds when the fruit splits and contracts, propelling the seeds through the air. The trajectory of these seeds can be determined with a high-speed camera. In an experiment on one type of plant, seeds are projected at 20 cm above ground level with initial speeds between 2.3 m/s and 4.6 m/s. The launch angle is measured from the horizontal, with $+90^\circ$ corresponding to an initial velocity straight up and -90° straight down.

3.82 The experiment is designed so that the seeds move no more than 0.20 mm between photographic frames. What minimum frame rate for the high-speed camera is needed to achieve this? (a) 250 frames/s; (b) 2500 frames/s; (c) 25,000 frames/s; (d) 250,000 frames/s.

3.83 About how long does it take a seed launched at 90° at the highest possible initial speed to reach its maximum height? Ignore air resistance. (a) 0.23 s; (b) 0.47 s; (c) 1.0 s; (d) 2.3 s.

3.84 If a seed is launched at an angle of 0° with the maximum initial speed, how far from the plant will it land? Ignore air resistance, and assume that the ground is flat. (a) 20 cm; (b) 93 cm; (c) 2.2 m; (d) 4.6 m.

3.85 A large number of seeds are observed, and their initial launch angles are recorded. The range of projection angles is found to be -51° to 75° , with a mean of 31° . Approximately 65% of the seeds are launched between 6° and 56° . (See W. J. Garrison et al., "Ballistic seed projection in two herbaceous species," *Amer. J. Bot.*, Sept. 2000, 87:9, 1257–64.) Which of these hypotheses is best supported by the data? Seeds are preferentially launched (a) at angles that maximize the height they travel above the plant; (b) at angles below the horizontal in order to drive the seeds into the ground with more force; (c) at angles that maximize the horizontal distance the seeds travel from the plant; (d) at angles that minimize the time the seeds spend exposed to the air.

ANSWERS

Chapter Opening Question ?

(iii) A cyclist going around a curve at constant speed has an acceleration directed toward the inside of the curve (see Section 3.2, especially Fig. 3.12a).

Key Example VARIATION Problems

VP3.9.1 (a) 1.53 s, 11.5 m (b) 3.06 s, 61.2 m

VP3.9.2 (a) 20.6 m/s (b) 5.40 m

VP3.9.3 (a) 3.51 s (b) 33.8 m (c) $v_x = 9.64$ m/s, $v_y = -22.9$ m/s

VP3.9.4 (a) $t = 2v_0/g$, $x = 2v_0^2/g$, $y = -2v_0^2/g$ (b) $t = v_0/g$, $x = v_0^2/g$, $y = -v_0^2/2g$

VP3.12.1 20.0 m

VP3.12.2 (a) 41.6 s (b) 6.04 m/s²

VP3.12.3 (a) P : 0.938 m/s, Q : 1.88 m/s (b) P : 8.79 m/s², Q : 17.6 m/s² (c) increase, increase

VP3.12.4 (a) Venus: 3.49×10^4 m/s, Earth: 2.99×10^4 m/s, Mars: 2.41×10^4 m/s (b) Venus: 1.13×10^{-2} m/s², Earth: 5.95×10^{-3} m/s², Mars: 2.55×10^{-3} m/s² (c) decrease, decrease

VP3.12.5 (a) $T_A/T_B = 4$ (b) $a_{\text{rad}, A}/a_{\text{rad}, B} = 1/8$

VP3.15.1 (a) +53.0 m/s (b) -53.0 m/s

VP3.15.2 (a) 90.0 m/s eastward (b) 90.0 m/s westward (c) neither

VP3.15.3 (a) 25.6 m/s, 51.3° north of east (b) 25.6 m/s, 51.3° south of west

VP3.15.4 130 m/s, 12.6° west of north

Bridging Problem

$$(a) R = \frac{2v_0^2 \cos(\theta + \phi) \sin \phi}{g \cos^2 \theta} \quad (b) \phi = 45^\circ - \frac{\theta}{2}$$